

SOP-TC820LTYsi-298 开箱、吊运和放置	
部件/系统 component	machine installation
安全等级 safety_level	critical
前置条件 preconditions	• 合格机器吊装人员执行 • 清理吊运路径并确认无人处于运动范围
作业步骤 steps	1. 拆解木托盘并取出附件。 2. 安装找平螺钉和外侧找平垫块。 3. 将叉车货叉穿过底座，确认伸过远侧并检查平衡。 4. 由第二人引导，缓慢放到找平垫块。 5. 拆出冷却液箱、排屑机和包装支架。
安全要求 safety_warnings	• 使用至少 15,000 lb 叉车 • 主轴箱侧较重，确认平衡 • 不得突然停止或提升过高
验收标准 acceptance_criteria	• 机器稳定、无倾覆和运输损伤。
关键词 keywords	15000 lb • 吊装 • 找平垫块
来源 page / page_end / section_code	手册 LTY-SOP • P59–P61
原文摘录 source_excerpt	3.14.2 TC820LTYsi Installation Checklist Installer – Use this checklist to assure a complete setup on the TC820LTYsi. Installation Checklist ITEMS TO BE COMPLETED BY QUALIFIED MACHINE RIGGER, IF NOT COMPLETED FOLLOW THE SUBSEQUENT STEPS: The Turning Center is packed on one wooden pallet during transportation: • When arriving at the machine the rigger should have carefully disassembled the wooden pallet and removed all the loose parts from the machine. • The leveling screws on the machine should be thread down by the rigger approximately 1.5” as measured from the bottom of the leveling screw to the bottom of the base casting. • The rigger should have placed the machine on the 4 outside leveling pads. • The coolant tank is packed separately on the pallet; the rigger should first remove any brackets holding the coolant tank in place, then remove the coolant tank from the pallet and set it aside. • The chip conveyor is packed separately on the pallet; the rigger should first remove any brackets holding the chip conveyor in place, then remove the chip conveyor from the pallet and install the casters on swarf conveyer before placing it in the machine casting. Refer to section 3.3 Uncrating the TCsi, and 3.4 Lifting and Placing the TCsi in the Service Manual. The machine should be lifted and moved by a fork-lift truck • Use a fork-lift truck with a minimum loading capacity of 15,000 lbs to lift and move the machine. • Place the forks of the fork-lift under the machine. • Make sure the forks of the fork-lift protrude past the far edge of the machine base and ensure the machine is balanced before starting to lift and move it. • Pay special attention to the machine balance while lifting and moving, please note that the weight of the machine is greater on the headstock side. • Do not stop the lifting motions suddenly during the process and prevent sudden movement as the machine could become unbalanced. • Do not raise the machine too high as this may create a loss of stability and cause the fork-lift truck to overturn. • Have another person to help guide the way while lifting and moving the machine. Ensure nobody is in the way and the path is not blocked before moving the machine. • When placing the machine on leveling pads be careful when removing machine from the forklift truck, the machine is subject to tipping over if placed even a little off balance. Please contact TRAK customer service and a qualified machine rigger if you plan to lift and move the machine using an overhead crane. Only certified equipment should be

SOP-TC820LTYsi-299 拆除运输固定支架	
部件/系统 component	shipping brackets
安全等级 safety_level	critical
前置条件 preconditions	• 机器稳定放置并可安全供电
作业步骤 steps	1. 拆前门安全开关附近固定板。 2. 从后方拆除 Z、X/Y 轴支架。 3. 拆除尾座支架。 4. 按手册步骤处理刀塔木块，不能直接手动取出。 5. 将支架和硬件装袋放入工具箱。
安全要求 safety_warnings	• 支架未拆前不得运行 • 不得手动移动自动尾座 • 支架和硬件不得丢弃
验收标准 acceptance_criteria	• 所有支架和木块已按步骤移除，支架已保存。
关键词 keywords	运输支架 • 碰撞 • 尾座
来源 page / page_end / section_code	手册 LTY-SOP • P40–P42
原文摘录 source_excerpt	3. Remove the bracket between the tailstock bed and tailstock base see figure 2.6.1c. Figure 2.6.1c Turret & Tailstock 4. Note the wooden block between the bed and the turret see figure 2.6.1c. DO NOT ATTEMPT TO REMOVE THE SUPPORT AT THIS TIME. Power needs to be supplied to move the X-Axis in the positive direction to remove the block. Make sure this is removed before using the machine! 3.6.2 TC820LTYsi Shipping Brackets On the TC820LTYsi machines there are shipping brackets that hold the axes in place during shipping. These brackets must be removed before operating the machine. Do not discard the shipping brackets. The shipping brackets must be replaced if the machine is ever moved.

SOP-TC820LTYsi-300 电气接线和接地检查	
部件/系统 component	electrical system
安全等级 safety_level	critical
前置条件 preconditions	• 合格电工执行 • 电源已隔离且配置与铭牌一致
作业步骤 steps	1. 确认 480 V 或变压器 208 V 配置。 2. 完成接地、进线密封和应力释放。 3. 按图纸核对进线和断路器。 4. 逐相测量并记录 456–504 V。 5. 用相序表确认顺时针相序。
安全要求 safety_warnings	• 三相 480 V; 432–528 V 可接受 • 现场核验 456–504 V • 约 8 ft 铜接地棒和 6 AWG 接地线
验收标准 acceptance_criteria	• 电压、相序、接地和进线防护符合要求。
关键词 keywords	480V • 接地 • 相序 • 456-504V
来源 page / page_end / section_code	手册 LTY-SOP • P42–P60
原文摘录 source_excerpt	3. Remove the bracket between the tailstock bed and tailstock base see figure 2.6.1c. Figure 2.6.1c Turret & Tailstock 4. Note the wooden block between the bed and the turret see figure 2.6.1c. DO NOT ATTEMPT TO REMOVE THE SUPPORT AT THIS TIME. Power needs to be supplied to move the X-Axis in the positive direction to remove the block. Make sure this is removed before using the machine! 3.7 Electrical Connection The TCsi machines come either with 208VAC input or 480VAC input. A 208VAC to 480VAC transformer is required for a 208VAC input. A 480VAC input does not require the transformer option. DANGER!!! Only a qualified electrician should wire the electricity for your machine installation. 3.7.1 Main Power to the Machine Connections The TCsi machine should be installed under the correct electrical environments. Before connecting the power wires, make sure the voltage is the same for both the machine and the plant power. Ensure the electrical cables have the same or better power rating as prescribed. Do not connect any power cable that might cause an abrupt voltage drop or generate signal noises on the power panel of the machine. Seal the electrical cabinet from moisture at the incoming power cable opening.

SOP-TC820LTYsi-301 机床找平和扭曲检查	
部件/系统 component	machine alignment
安全等级 safety_level	high
前置条件 preconditions	• 运输支架已拆除 • 找平垫块稳定 • 使用校准水平仪
作业步骤 steps	1. 从 Z 轴箱式导轨两侧测量。 2. 将刀塔置于中心并逐点调整找平螺钉。 3. 沿全 Z 行程检查 X/Z 扭曲。 4. 锁紧螺母且不改变螺钉高度。
安全要求 safety_warnings	• 水平 ≤ 0.0006 in/12 in • X/Z 扭曲 < 0.0006 in/12 in
验收标准 acceptance_criteria	• 多位置重复测量后仍在阈值内。
关键词 keywords	找平 • 扭曲 • 0.0006 in
来源 page / page_end / section_code	手册 LTY-SOP • P45–P47
原文摘录 source_excerpt	208 T BLACK WHITE L1 INCOMING S FAN FAN L2 L3 POWER R 480V L1 L2 L3 POWER TO MACHINE Figure 2.7.4 Transformer Wiring Configuration 220 Volt Options 208 T L1 L2 INCOMING S L3 480V POWER R BLACK WHITE POWER TO FAN FAN MACHINE L1 L2 L3 Figure 2.7.5 Transformer Wiring Configuration 240 Volt Options 3.8 Machine Leveling 3.8.1 Leveling Procedure The leveling of the TCsi machine consists of adjusting leveling screws (F1 – F4 and R1 – R4), shown in figure 2.8.1a, such that the machine is level using either analog or digital instruments. If the machine is not leveled properly, it may become twisted. Even a slight amount of twist will move centers out of alignment, and result in inaccurate work.

SOP-TC820LTYsi-302 冷却液箱、冷却泵和冲屑泵安装	
部件/系统 component	coolant system
安全等级 safety_level	high
前置条件 preconditions	• 机床已找平 • 泵和接线处于断电状态
作业步骤 steps	1. 将冷却液箱滑入排屑机下方/右侧。 2. 普通泵接 U3, chip wash pump 接 U4。 3. 接软管并检查密封。 4. 从 MCP 启动两泵，核对标签旋向。 5. 确认冷却流经刀塔、冲屑流经 hose 且无泄漏。
安全要求 safety_warnings	• TC820LTYsi 使用 50 gallon 冷却液箱 • 普通泵接 U3、冲屑泵接 U4 • 旋向错误或泄漏不得运行
验收标准 acceptance_criteria	• U3/U4 接线、旋向、流量和泄漏检查通过。
关键词 keywords	50 gallon • U3 • U4 • chip wash
来源 page / page_end / section_code	手册 LTY-SOP • P47–P48
原文摘录 source_excerpt	• Once the machine is sufficiently leveled lower screws F2, R2, F3 and R3 onto their leveling pads by hand and use a wrench and tighten only by 10°. Do not tighten any further as you will change the level of the machine. • Verify that the machine is still level by checking for twist at different points along the Z-Axis Box Ways. • Tighten the locking nuts for all leveling screws making sure that you do not raise or lower any of the leveling screws. • Verify that the machine is still level by checking for twist at different points along the Z-Axis Box Ways. Further Checks • To further check for twist in the machine a leveling plate can be attached to the turret with two calibrated bubble type levels or a multi axis digital level placed level on the plate, shown in figure 2.8.1.c. • Starting at the spindle of the machine, move the turret along the Z-Axis, stopping at 4 positions and allowing the bubble to settle. The change in level must be within 0.002" /10" . Figure 2.8.1c Leveling Plate 3.9 Placing the Coolant System The TC820si machine is fitted with a 33-gallon coolant tank and pump that delivers coolant through the turret at 30psi. The TC820LTYsi machine is fitted with a 50-gallon coolant tank and pump that delivers coolant through the turret at 70psi. Once the machine is leveled the coolant tank can be placed. For the TC820si machine, with the pump removed starting from the front of the machine, push the coolant tank through the center of the machine' s base casting. The front face of the coolant tank should sit flush with the sheet metal guarding. For the TC820LTYsi machine, the coolant tank can slide in the right-hand side of the machine under the chip conveyor with the pumps located at the back of the machine.

SOP-TC820LTYsi-303 排屑机安装和测试	
部件/系统 component	chip conveyor
安全等级 safety_level	high
前置条件 preconditions	- 脚轮已安装 - 排屑机位于铸件内且运动区无人
作业步骤 steps	- 安装脚轮并放置排屑机。 - 电机电缆接 U6并做应力释放。 - 用 MCP 测正转和反转。 - 安装盖板，使冷却液流入排屑机。
安全要求 safety_warnings	- 电缆接 U6 - 正反方向正确 - 反转按住 MCP 按键
验收标准 acceptance_criteria	正反转、回流和电缆布置正常。
关键词 keywords	U6 • 排屑机 • 正转 • 反转
来源 page / page_end / section_code	手册 LTY-SOP • P49–P49
原文摘录 source_excerpt	3.10 Setting Up Chip Conveyor The chip conveyor will be delivered installed in the machine envelope, make sure to install the wheels before placing the machine on the ground, see figure 2.10a. The chip conveyor is used to evacuate chips during machining. Connect the cable from the chip conveyor motor to U6 on the electrical cabinet side facing the coolant pump see figure 2.10b. To test the chip conveyor, use the forward and reverse buttons on the MCP and confirm the directions are correct. Figure 2.10a Chip Conveyor TCsi Figure 2.10b Chip Conveyor Connection TC820si Chip Conveyor Connection TC820LTYsi

SOP-TC820LTYsi-304 卡盘和尾座套筒脚踏开关	
部件/系统 component	foot switches
安全等级 safety_level	high
前置条件 preconditions	- 脚踏开关未接电 - 操作区域清洁
作业步骤 steps	- 将两个脚踏开关放在机器前方。 - 沿左侧布线并连接电柜下方连接器。 - 打开门测试卡盘夹紧/松开。 - 测试尾座套筒前进/后退。 - 核对卡盘 255 psi、限压 425 psi、套筒 70/140 psi。
安全要求 safety_warnings	- 测试卡盘时人员远离卡爪 - 电缆远离地面和运动件
验收标准 acceptance_criteria	脚踏开关方向正确，压力符合安装要求。
关键词 keywords	脚踏开关 • 255 psi • 70 psi • 尾座套筒
来源 page / page_end / section_code	手册 LTY-SOP • P50–P50
原文摘录 source_excerpt	3.11 Setting Up Foot Switches 3.11.1 Setting Up Chuck Foot Switch The chuck foot switch will be delivered in a box along with the TCsi machine. Place the foot pedal in a convenient position for the operator in the front of the machine see figure 2.11a. Route the cable under the left side of the machine to the connector under the electrical cabinet close to the rear door. The connector only attaches one way, and the location is labeled inside the cabinet see figure 2.11b, inset photo installed cable under cabinet. To test its operation, with the door open press the foot pedal and the chuck jaws should actuate. Figure 2.11.1a Chuck Foot Switch Figure 2.11.1b Switch Connection 3.11.2 Setting Up Quill Foot Switch On the TC820LTYsi machines the quill foot switch will be delivered in a box along with the machine. Place the foot pedal in a convenient position for the operator in the front of the machine see figure 2.11.2a. Route the cable under the left side of the machine to the connector under the electrical cabinet close to the rear door. The connector only attaches one way, and the location is labeled inside the cabinet see figure 2.11.2b, inset photo installed cable under cabinet. To test its operation, press the foot pedal forward and the tailstock quill should move forward and press the foot pedal back and the tailstock quill should move back towards its original position. Figure 2.11.2a Quill Foot Switch Figure 2.11.2b Switch Connection

SOP-TC820LTYsi-305 液压和润滑系统调试	
部件/系统 component	hydraulic and lubrication
安全等级 safety_level	high
前置条件 preconditions	• 完成接线、找平和支架拆除 • 液压/润滑油箱可安全检查
作业步骤 steps	1. 释放急停并按 Reset 启动液压泵。 2. 检查液压油位、ISO VG32 和泄漏。 3. 按 MCP 润滑键 5 次运行润滑泵。 4. 确认润滑压力 70 psi。 5. 点动 X/Y/Z，确认导轨和丝杠见油。
安全要求 safety_warnings	• 压力异常立即停机 • 保持远离运动件和高压油路
验收标准 acceptance_criteria	• 液压稳定、无泄漏、润滑到达导轨和丝杠。
关键词 keywords	700 psi • 70 psi • ISO VG32 • ISO VG68
来源 page / page_end / section_code	手册 LTY-SOP • P61–P62
原文摘录 source_excerpt	Installation Checklist (8) Cast Iron Leveling Pads (May already be placed under the machine) (2) Sets of 3 pendant keys 8 Verify 2 electrical cabinet keys are zip tied to the electrical cabinet door lock on the • inside of the electrical cabinet. Remove covers and shipping brackets: • On the TC820LTYsi machines there are shipping brackets that hold the axes in place during shipping, these brackets must be removed before operating the machine, see section 2.6.2 of the Service Manual. • Remove the sheet metal cover on rear of the machine exposing the Z-9 axis box ways and the X-axis and the Y-axis casting. • Remove the Z-axis shipping bracket. • Remove the X/Y-axis shipping bracket. • Remove the sheet metal cover on the tailstock side of the machine exposing the rear of the tailstock box ways. • Remove the tailstock shipping bracket. • DO NOT MOVE THE TAILSTOCK BY HAND, THIS WILL CAUSE IT TO LOSE POSITION AND MAY CAUSE A CRASH. • Remove the front door shipping bracket. • Place the shipping brackets and hardware in a bag and store in the machine toolbox. 10 Clean the machine if needed and remove any remaining protective grease. Turn on the power to the machine. To start machine after boot up: • Release all E-Stop buttons. • Press and hold reset button, at this point you should hear the hydraulic pump • and system start (the pump should rotate CCW per the label on the pump). • Turn Feed Start on to allow axis motion. • Turn Spindle Start on to allow spindle motion. Check the level of the machine from both sides of the Z-axis box ways. Center the turret 12 as required before leveling. The machine should be level to within 0.0006"/12" . Refer to section 3.8 of the Service Manual. 13 Check the X and Z-axis twist from the turret across the entire Z-axis travel. The machine • twist should be < 0.0006" /12" . 14 Confirm the MCP panel has the key plugged in to change the access level. Key is • shipped in toolbox. Check the Hydraulic Fluid Tank level and fill with ISO VG 32 oil if needed. • Inspect all hydraulic lines, fittings and components to make sure there are no leaks. 15 Run the lube pump by pressing the lube button on MCP 5 times.

SOP-TC820LTYsi-306 开机验收、TAILSTOCK_TEST 和 TURRET_TEST	
部件/系统 component	machine acceptance
安全等级 safety_level	critical
前置条件 preconditions	• 安装和安全检查完成 • 人员远离运动区
作业步骤 steps	1. 测试 MPG、X/Y/Z 方向、软件限位、急停、门互锁、工作灯和主轴制动。 2. 测试尾座销、停车开关、工件开关和限位。 3. 运行 TAILSTOCK_TEST 15 分钟。 4. 运行 TURRET_TEST，验证正反分度和刀具表工位。 5. 测试探头、主轴 100–4000 RPM 和动力刀具 100–4500 RPM。
安全要求 safety_warnings	• 急停、门互锁、主轴制动失败不得生产 • 持续观察噪声、泄漏和状态灯
验收标准 acceptance_criteria	• 安全功能、尾座、刀塔、探头、主轴和动力刀具均有记录。
关键词 keywords	TAILSTOCK_TEST • TURRET_TEST • 100-4000 RPM • 4500 RPM
来源 page / page_end / section_code	手册 LTY-SOP • P64–P66
原文摘录 source_excerpt	Installation Checklist • Inspect all coolant lines, fittings and components to make sure there are no leaks. Check spindle motor cooling: • Remove the sheet metal cover to the spindle motor fan • Make sure the air is blowing towards the motor. Check electrical cabinet cooling: • Open the electrical cabinet door. • Make sure the air is blowing into the heat exchange on both sides of the door. • Make sure the cabinet circulation fan is blowing towards the components in the electrical cabinet. Check MPG functions correctly: • Move all axis at all speed settings and confirm each axis moves in the correct 22 direction. • Confirm Deadman switch on MPG is working to move the axis when the door is open. Test machine MCP motion and limits: • Set the MPG to off. • Test each axis by using the Axis + and - button with and without the RAPID 23 button on the MCP. • Verify the machine stops on the software limits before hitting the hard stop. Adjust parameters 36100 (-) and 36110 (+) on the axis in question if you do hit a hard stop. Check to make sure that the E-Stop button is functioning correctly: • Turn spindle on and jog an axis and press the E-stop button during this operation and verify the spindle and axis stops. Test door interlock: • Close the door and make sure the control recognizes the door as being closed. • Once you close the door, you should not be able to open it until you press the door open button on the MCP.

SOP-TC820LTYsi-307 自动尾座参考位置恢复	
部件/系统 component	automatic tailstock
安全等级 safety_level	critical
前置条件 preconditions	• 确认自动尾座丢失位置 • 尾座和 Z 轴无干涉
作业步骤 steps	1. 移动 Z 轴使尾座销对齐 pickup hole。 2. 用 MCP 松开尾座。 3. 用 MCP 使尾座销啮合开关。 4. 退回尾座销并夹紧尾座。
安全要求 safety_warnings	• 禁止手动推动尾座 • 尾座销与 pickup hole 对齐后操作
验收标准 acceptance_criteria	• 控制器重新学习尾座相对 Z 轴位置。
关键词 keywords	pickup hole • 参考位置 • 尾座销
来源 page / page_end / section_code	手册 LTY-SOP • P117–P117
原文摘录 source_excerpt	7.3.1 Automatic Tailstock Has Lost its Position It is important that the TC820LTYsi tailstock is not moved by hand. If the tailstock is moved by hand the control will lose its position and will not be able to locate it for automatic tailstock commands. If the control loses the position of the automatic tailstock: • Move the Z-axis to line the tailstock pin up with the pickup hole on the Z-Axis. There are markers on the Z-Axis and the Tailstock body that show when this is lined up. • Unclamp the tailstock using the button on the MCP. • Engage the tailstock pin using the button on the MCP. The pin must engage the switch in the pickup hole to learn the new position. • Disengage the tailstock pin using the button on the MCP. • Clamp the tailstock using the button on the MCP. The machine should now have learned the position of the tailstock relative to the z-axis machine coordinate system. 7.3.2 Automatic Tailstock in JOG Mode To correctly position the tailstock manually in jog mode: • Move the Z-axis to line the tailstock pin up with either the park hole or pick up hole (depending on where they want to move the tailstock to). • Unclamp the tailstock using the button on MCP button or M code. • Engage the tailstock pin using the MCP button or M code. • Move the Z-axis to position the tailstock in the location desired. • Disengage the tailstock pin using the MCP button or M code. • Clamp the tailstock using the MCP button or M code. - o The tailstock location can be found from the z-axis position in the machine coordinate system or from the custom screen (shown below) in the programing environment that can be used for automatically positioning the tailstock in a part program. • Set the quill pressure. • Position the quill in the desired position. • Open the tailstock cover to move the quill sensor to match the required position. - o The user would do this to set the tailstock position before running an automatic cycle. From this they will have the quill position set and a tailstock location they can use in their part program. • Close the tailstock cover to make the machine ready for machining. • The tailstock can be manually put on the park switch following the first 6 steps of the method outlined above and moving it back onto the park switch. • The tailstock can automatically be moved to the park switch using a m-code. 7.3.3 Automatic Tailstock in Auto Mode To correctly position the tailstock manually in auto mode: If the user wants to automatically move the tailstock in a part program, they must do it by using the custom screen. For this the tailstock position must be known by the machine and the user must know where they want to position the tailstock relative to the Z-axis machine coordinate system. To use the automatic tailstock cycle in a part program the user must: • Go to the Tailstock custom screen in the part programming environment. o The tailstock custom screen can be found by pressing the tailstock button on the second page of the horizontal soft keys. • ZP: New Tailstock Position

SOP-TC820LTYsi-308 自动尾座 JOG 模式定位	
部件/系统 component	automatic tailstock
安全等级 safety_level	high
前置条件 preconditions	• 主轴停止且路径清空 • 确定 park hole 或 pickup hole
作业步骤 steps	1. 对齐目标孔位。 2. 松开尾座并啮合尾座销。 3. 移动 Z 轴到目标位置。 4. 退销并夹紧。 5. 设置套筒压力和位置，必要时调整传感器。
安全要求 safety_warnings	• 不允许手动推移 • 啮合/退销前不得移动 Z 轴
验收标准 acceptance_criteria	• 尾座位置、park/part switch 与实际状态一致。
关键词 keywords	JOG • park hole • pickup hole
来源 page / page_end / section_code	手册 LTY-SOP • P117–P117
原文摘录 source_excerpt	7.3.1 Automatic Tailstock Has Lost its Position It is important that the TC820LTYsi tailstock is not moved by hand. If the tailstock is moved by hand the control will lose its position and will not be able to locate it for automatic tailstock commands. If the control loses the position of the automatic tailstock: • Move the Z-axis to line the tailstock pin up with the pickup hole on the Z-Axis. There are markers on the Z-Axis and the Tailstock body that show when this is lined up. • Unclamp the tailstock using the button on the MCP. • Engage the tailstock pin using the button on the MCP. The pin must engage the switch in the pickup hole to learn the new position. • Disengage the tailstock pin using the button on the MCP. • Clamp the tailstock using the button on the MCP. The machine should now have learned the position of the tailstock relative to the z-axis machine coordinate system. 7.3.2 Automatic Tailstock in JOG Mode To correctly position the tailstock manually in jog mode: • Move the Z-axis to line the tailstock pin up with either the park hole or pick up hole (depending on where they want to move the tailstock to). • Unclamp the tailstock using the button on MCP button or M code. • Engage the tailstock pin using the MCP button or M code. • Move the Z-axis to position the tailstock in the location desired. • Disengage the tailstock pin using the MCP button or M code. • Clamp the tailstock using the MCP button or M code. - o The tailstock location can be found from the z-axis position in the machine coordinate system or from the custom screen (shown below) in the programing environment that can be used for automatically positioning the tailstock in a part program. • Set the quill pressure. • Position the quill in the desired position. • Open the tailstock cover to move the quill sensor to match the required position. - o The user would do this to set the tailstock position before running an automatic cycle. From this they will have the quill position set and a tailstock location they can use in their part program. • Close the tailstock cover to make the machine ready for machining. • The tailstock can be manually put on the park switch following the first 6 steps of the method outlined above and moving it back onto the park switch. • The tailstock can automatically be moved to the park switch using a m-code. 7.3.3 Automatic Tailstock in Auto Mode To correctly position the tailstock manually in auto mode: If the user wants to automatically move the tailstock in a part program, they must do it by using the custom screen. For this the tailstock position must be known by the machine and the user must know where they want to position the tailstock relative to the Z-axis machine coordinate system. To use the automatic tailstock cycle in a part program the user must: • Go to the Tailstock custom screen in the part programming environment. - o The tailstock custom screen can be found by pressing the tailstock button on the second page of the horizontal soft keys. • ZP: New Tailstock Position

SOP-TC820LTYsi-309 自动尾座 AUTO 模式定位	
部件/系统 component	automatic tailstock
安全等级 safety_level	high
前置条件 preconditions	• 控制器已知尾座参考位置 • 首件已确定 Z 位置和套筒位置
作业步骤 steps	1. 进入 Tailstock custom screen。 2. 输入 ZP 目标位置。 3. 在 QL 选择 Forward 或 Back。 4. 确认 ZPOS 当前值并选择 Accept。 5. 需要停车时使用手册 M-Code。
安全要求 safety_warnings	• 验证 ZP/QL，避免进入工件或刀具路径 • 停车前确认空间和限位逻辑
验收标准 acceptance_criteria	• 程序尾座定位与首件设置一致。
关键词 keywords	AUTO • ZP • QL • ZPOS
来源 page / page_end / section_code	手册 LTY-SOP • P117–P118
原文摘录 source_excerpt	7.3.1 Automatic Tailstock Has Lost its Position It is important that the TC820LTYsi tailstock is not moved by hand. If the tailstock is moved by hand the control will lose its position and will not be able to locate it for automatic tailstock commands. If the control loses the position of the automatic tailstock: • Move the Z-axis to line the tailstock pin up with the pickup hole on the Z-Axis. There are markers on the Z-Axis and the Tailstock body that show when this is lined up. • Unclamp the tailstock using the button on the MCP. • Engage the tailstock pin using the button on the MCP. The pin must engage the switch in the pickup hole to learn the new position. • Disengage the tailstock pin using the button on the MCP. • Clamp the tailstock using the button on the MCP. The machine should now have learned the position of the tailstock relative to the z-axis machine coordinate system. 7.3.2 Automatic Tailstock in JOG Mode To correctly position the tailstock manually in jog mode: • Move the Z-axis to line the tailstock pin up with either the park hole or pick up hole (depending on where they want to move the tailstock to). • Unclamp the tailstock using the button on MCP button or M code. • Engage the tailstock pin using the MCP button or M code. • Move the Z-axis to position the tailstock in the location desired. • Disengage the tailstock pin using the MCP button or M code. • Clamp the tailstock using the MCP button or M code. - o The tailstock location can be found from the z-axis position in the machine coordinate system or from the custom screen (shown below) in the programing environment that can be used for automatically positioning the tailstock in a part program. • Set the quill pressure. • Position the quill in the desired position. • Open the tailstock cover to move the quill sensor to match the required position. - o The user would do this to set the tailstock position before running an automatic cycle. From this they will have the quill position set and a tailstock location they can use in their part program. • Close the tailstock cover to make the machine ready for machining. • The tailstock can be manually put on the park switch following the first 6 steps of the method outlined above and moving it back onto the park switch. • The tailstock can automatically be moved to the park switch using a m-code. 7.3.3 Automatic Tailstock in Auto Mode To correctly position the tailstock manually in auto mode: If the user wants to automatically move the tailstock in a part program, they must do it by using the custom screen. For this the tailstock position must be known by the machine and the user must know where they want to position the tailstock relative to the Z-axis machine coordinate system. To use the automatic tailstock cycle in a part program the user must: • Go to the Tailstock custom screen in the part programming environment. - o The tailstock custom screen can be found by pressing the tailstock button on the second page of the horizontal soft keys. • ZP: New Tailstock Position

SOP-TC820LTYsi-310 尾座套筒传感器定位	
部件/系统 component	tailstock sensors
安全等级 safety_level	high
前置条件 preconditions	• 设备停止且确认需要调整传感器
作业步骤 steps	1. 确认 700427 或相关位置消息。 2. 清洁并拆下四个罩螺钉。 3. 松开六角螺母滑动传感器座。 4. 使指示灯在正确位置点亮并紧固。 5. 整理线缆、清洁垫片并复装。
安全要求 safety_warnings	• 清洁后拆罩，防止切屑进入 • 保护垫片和线缆
验收标准 acceptance_criteria	• 前进/退回反馈与实际位置一致。
关键词 keywords	700427 • 接近开关 • 传感器
来源 page / page_end / section_code	手册 LTY-SOP • P119–P120
原文摘录 source_excerpt	7.3.4 Extended Tail Stock Will Not Respond to Control Sensors inside the tailstock body can be used to determine the tailstock quill position and motion during operation. In order to set the tailstock quill, position the operator will have to adjust a sensor under the tailstock cover. See figure 6.3.2a showing the tailstock position indicator and the control measurement sensors. For the control to allow a cycle start command the tailstock quill must be sat on its backward sensor. The control will display the message, “700427 Tailstock Quill is no located on the Advanced or retracted Proximity Switch” . Figure 6.3.2a shows the details of the forward and backward sensors which need positioning. Figure 6.3.2a Tailstock Sensors In order to set the tailstock quill position sensors, first clean off the cover and surrounding areas to prevent dirt and debris from entering tailstock. In order to position the sensors with the tailstock quill, remove the four knurled screws that hold the cover in position see figure 6.3.2b. Be careful with the gasket, it needs to be cleaned and reused. Figure 6.3.2b Tailstock Sensor Access Cover

SOP-TC820LTYsi-311 尾座中心线、Z 轴和 B 轴对中	
部件/系统 component	tailstock alignment
安全等级 safety_level	high
前置条件 preconditions	• 碰撞或加工锥度时执行 • 测量面清洁
作业步骤 steps	1. 用千分表扫描尾座套筒孔，中心线误差 ≤ 0.002 in。 2. 用 MT4 对中杆和磁性表座检查 Z 平行度。 3. 顶部/侧面误差 < 0.001 in/4 in。 4. B 轴移到另一端重复检查。 5. 超差时用调整螺栓小量修正。
安全要求 safety_warnings	• 调整前必须有精密测量设备 • 小量调整并复测
验收标准 acceptance_criteria	• 中心线、Z 和 B 轴平行度达到阈值。
关键词 keywords	MT4 • 0.002 in • 0.001 in/4 in
来源 page / page_end / section_code	手册 LTY-SOP • P115–P116
原文摘录 source_excerpt	7.2 Tailstock Alignment 7.2.1 Tailstock to Spindle Centerline Alignment The machine should have a common centerline between the spindle, turret pocket at the X0 position, and the tailstock. After a crash occurs the spindle to tailstock centerline should be checked and adjusted accordingly. The maximum permitted error is 0.002" (0.05mm). To Check the current spindle to turret pocket centerline alignment: 1. Clamp an indicator in the chuck and move the tailstock so the indicator feeler is measuring the pocket of the tailstock quill. 2. Make sure that the measuring surfaces are clean. Any debris or dirt will affect the measurements. 3. Sweep the pocket of the tailstock quill and measure the error in the pocket. The error must not exceed 0.002" (0.05mm). 4. If this value is exceeded you will need to adjust your tailstock centerline. 7.2.2 Tailstock to Z-Axis Parallelism It is important when using the tailstock, it is aligned to the Z-Axis motion on the machine. To check the tailstock parallelism: 1. Install a MT4 taper alignment tool into the tailstock quill. 2. Attach a magnetic base to the turret body. 3. Set up your indicator so that it is reading off the side of the alignment bar. Be sure to have your indicator set up in a way to reduce any cosine errors present in the reading. 4. Make sure that the measuring surfaces are clean. Any debris or dirt will affect the measurements. 5. Move the turret in the Z direction so your indicator runs across the top of the alignment bar. 6. Record the difference in position on your indicator from the start to the end of travel along the alignment bar. 7. Repeat this process taking a reading from the top of the alignment bar. 8. The error along the top and the side of the alignment bar should be less than 0.001" /4" (0.025mm/100mm). 9. If the indicator shows an error of greater than this your tailstock needs realigning to make it move parallel to the Z-Axis motion in the machine. NOTE: if you do not have a MT4 alignment bar, place a precision ground bar between centers on the headstock and tailstock measuring the error along the top and side of the precision ground bar, see figure 6.2.2. The measured error across the span should be $< 0.001" /4"$. Figure 6.2.2 Tailstock Alignment Between Centers

SOP-TC820LTYsi-312 刀塔工位重新参考和同步	
部件/系统 component	turret
安全等级 safety_level	high
前置条件 preconditions	• 实际工位与刀具表不一致 • 刀塔周围无干涉
作业步骤 steps	1. MCP 按 JOG 和 REF PT。 2. 控制器进入 turret menu 执行 Turret Reference。 3. Feed Start 快按三次进入 service mode。 4. 查看 IP/OP Clamp 状态，必要时用 T REF/T PLUS/T Minus。 5. 退出前确认夹紧且工位一致。
安全要求 safety_warnings	• 进入 service mode 前确认主轴/尾座安全 • 退出前确认刀塔夹紧
验收标准 acceptance_criteria	• 刀塔工位与刀具表绿色工位一致。
关键词 keywords	T REF • REF PT • 刀塔参考
来源 page / page_end / section_code	手册 LTY-SOP • P106–P107
原文摘录 source_excerpt	6.0 Sauter 12 Station Turret 6.1 12 Station Hydraulic Turret The TCsi series is fitted with a 12-station turret from Sauter. The TC820si uses and Bolt-on Turret face and the TC820LTYsi is fitted with a BMT45 Turret face with a live tooling drive. The BMT45 turret for the TC820LTY is available in a half index option providing 24 indexing positions. The turret is driven by an AC servo motor and utilizes a hydraulic clamping mechanism and proximity switch. As the tool exchange command is issued, turret will be rotated to the selected tool position according to NC or manual commands. When the selected tool traces the target location, the AC servo motor will stop immediately. The hydraulic system will then lock the tool magazine to prevent the tool magazine from further traveling. The proximity switch is used for checking the magazine' s clamping. The rotation direction could be either cw or ccw. The tool is selected based on the shortest path to minimize the tool selection time. 6.2 Troubleshooting the 12 Station Turret 6.2.1 12 Station Turret Out of Sync with Control Display Occasionally the turret position will become out-of-synch with the control display. Typically, this is when the green highlighted entry of the Tool List as shown in figure 5.2.1a shows a different number than the tool position number engraved into the turret body at the current cutting position. Figure 5.2.1a Tool List Showing Tool Position To re-sync the turret with the tool table, on the MCP press “JOG” and “REF PT” . On the control press 1 - “>” arrow to pull out hidden window, 2 – turret menu, then 3 – Turret Reference as shown on figure 5.2.1b.

SOP-TC820LTYsi-313 刀塔 X/Z 轴平行度检查	
部件/系统 component	turret alignment
安全等级 safety_level	high
前置条件 preconditions	- 碰撞后或精度异常时执行 - 测量面清洁
作业步骤 steps	- 磁性表座安装于尾座导轨。 - 沿 X 轴约 4 in 扫测刀塔面。 - 沿 Z 轴约 4 in 扫测周边面。 - 两轴误差均需 < 0.0004 in/4 in。 - 按手册调整后交替复测。
安全要求 safety_warnings	- 调整两个螺钉会同时影响 X/Z - 超差由精密测量人员处理
验收标准 acceptance_criteria	X/Z 平行度均达标。
关键词 keywords	0.0004 in • X轴 • Z轴
来源 page / page_end / section_code	手册 LTY-SOP • P108–P110
原文摘录 source_excerpt	6.2.2 Turret Parallelism to Axis Motion It is essential that the turret is set up and aligned parallel to the axis motion of the machine in order to ensure proper accuracy. The turret parallelism to the axis motion should be checked after a crash has occurred, see figure 5.2.2. Figure 5.2.2 Ready to Measure 6.2.2.1 Turret to X-Axis Parallelism To check the turret parallelism to the X-Axis motion: 1. Attach a magnetic base to a tailstock way, see figure 5.2.2. 2. Set up your indicator so that it is reading off the turret's face where a turning tool will be mounted, see figure 5.2.2.1. Be sure to have your indicator set up in a way to reduce any cosine errors present in the reading. 3. Make sure that the measuring surfaces on the turret are clean. Any debris or dirt on the turret will affect the measurements. 4. Move the turret in the X direction so your indicator runs across its face. You should be able to achieve about 4 inches of travel. 5. Record the difference in position on your indicator from the start to the end of travel along the turret face. 6. The error should be less than 0.0004" /4" (0.010mm/100mm). 7. If the indicator shows an error of greater than this your turret needs realigning to make it move parallel to the X-Axis motion in the machine.

SOP-TC820LTYsi-314 主轴与刀塔中心线同心度	
部件/系统 component	spindle and turret alignment
安全等级 safety_level	high
前置条件 preconditions	- 碰撞后或加工异常时执行 - 刀塔和测量面清洁
作业步骤 steps	- 安装镗刀杆夹持座并移至 X0。 - 卡盘夹表，扫测夹持座孔。 - 按 $\pm X/\pm Y$ 位置记录总误差。 - 确认总误差 ≤ 0.002 in，超差则调整并复测。
安全要求 safety_warnings	- 卡盘夹表确认夹持可靠 - 调整由精密测量人员执行
验收标准 acceptance_criteria	刀塔孔相对主轴中心线总误差达标。
关键词 keywords	0.002 in • 同心度 • 碰撞
来源 page / page_end / section_code	手册 LTY-SOP • P111–P111
原文摘录 source_excerpt	The machine should have a common centerline between the spindle, turret pocket at the X0 position, and the tailstock. After a crash occurs the spindle to turret pocket centerline should be checked and adjusted accordingly. The maximum permitted error is 0.002" (0.05mm). To Check the current spindle to turret pocket centerline alignment: - Install a boring bar holder and move the turret so it is at X0,. - Make sure that the mating surfaces of the turret and tooling are clean. Any debris between the turret and the boring bar holder will affect the measurements. - Clamp an indicator in the chuck and move the turret so the indicator feeler is measuring the pocket of the boring bar holder. - Set your indicator zero at position 1, +X. Refer to figure 5.2.2.4a to show the positions of $\pm X$ and $\pm Y$. - Sweep the pocket of the boring bar holder and measure the error in the pocket. Refer to figure 5.2.2.4b to show the actual tooling used. - The total error must not exceed 0.002" (0.05mm). - If the total error exceeds this value you will need to adjust your turret centerline position. Figure 5.2.2.4a Spindle to Turret Centerline Figure 5.2.2.4b Tooling for Spindle to Turret Centerline 6.2.2.5 TC820LTY Driven Tooling Drive Alignment.

SOP-TC820LTYsi-315 动力刀具驱动对中

部件/系统 component	live tooling
安全等级 safety_level	critical
前置条件 preconditions	• 动力刀具不能正确啮合 • 移除动力刀具和前盖
作业步骤 steps	1. 用千分表检查两个耦合标记点。 2. 超差时拆 plug screw 并松开夹紧块螺钉。 3. 旋转耦合头使差值 < 0.0008 in。 4. 按 20、40、60 ft-lbs 逐级紧固并复测。 5. 复装 plug screw、动力刀具和前盖。
安全要求 safety_warnings	• 保持运动区无人 • 夹紧螺钉按 20/40/60 ft-lbs 逐级紧固
验收标准 acceptance_criteria	• 动力刀具完全啮合，差值 < 0.0008 in。
关键词 keywords	BMT45 • 0.0008 in • 60 ft-lbs
来源 page / page_end / section_code	手册 LTY-SOP • P112–P112
原文摘录 source_excerpt	If the driven tools are not properly engaging with the driven tooling drive check turret to live tooling drive alignment: • Remove all driven tools and the front cover of 5.2.2.5a BMT45 Live Tooling Gear Head the turret. Alignment Markers • Take the machine out of E-Stop and with an 5.2.2.5b BMT45 Plug Screw indicator check the alignment of the 2 marking points on the live tooling gear head coupling 5.2.2.5c BMT45 Gear Head Clamping Screw along the X-Axis, shown in figure 5.2.2.5a. • The level of the 2 marking points along the X- Axis direction should be within 0.0008in or 0.2mm. If the level is greater than 0.0008” or 0.2mm along the X-Axis direction the gear head coupling will need to be adjusted. • If the level is out of spec remove the plug screw on the side of the turret housing, shown in figure 5.2.2.5b. • Loosen the screw of the shaft clamping block from the hole opened by removing the plug screw from the side of the turret housing, figure 5.2.2.5c. The Gear coupling head should now be able to rotate. o NOTE: This screw should be torqued to 60 ft-lbs. When tightening ensure it is tightened to 60 ft-lbs. • Rotate the gear head coupling and sweep the indicator checking the 2 markings on the gear head coupling until they are level across the X- Axis within 0.0008” or 0.2mm. • Fastening the clamping block screw to 20 ft-lbs then 40lbs and finally 60 ft-lbs checking the marking level remains less than 0.0008” or 0.2mm after each tighten. • When the clamping block screw is tightened to 60 ft-lbs and the 2 markings level remains less than 0.0008” or 0.2mm reinstall the plug screw on the side of the turret housing. • Install a driven tool and verify the drive is fully engaged. • Reinstall the front cover of the tool disc and tighten the M6 bolts to 14 ft-lbs.

SOP-TC820LTYsi-316 探头校准前的刀具表设置

部件/系统 component	automatic tool setter probe
安全等级 safety_level	high
前置条件 preconditions	• 准备 1/3/5 号校准刀具 • 相邻刀具无干涉
作业步骤 steps	1. 工位 1 装 OD 粗车刀, 3 装 ID 粗车刀, 5 装切断刀。 2. 在 Tool List 新建并填写刀具数据。 3. 从 T,S,M 清除工作偏置, 选择 G500。
安全要求 safety_warnings	• 清除工作偏置后建立机器零点 • 确认刀具不会撞击探头/卡盘
验收标准 acceptance_criteria	• 三把刀具数据完整且 G500 生效。
关键词 keywords	Tool List • G500 • 工位1/3/5
来源 page / page_end / section_code	手册 LTY-SOP • P122–P123
原文摘录 source_excerpt	8.0 Tool Setter Probe 8.1 About Probe 8.1.1 Manual Tool Setting Arm and Probe A manual tool setting arm and probe is fitted to the TC820si machine as standard. The tool setting arm is manually deployed by the user and can be used in the jog environment to set tool data. The position of the tool setting arm and probe are stored in the machine and used to set the position of the tools cutting edge with reference to the work piece and machine coordinate system. If the tool setting arm or probe is ever crashed, it will require recalibration. Although your probe will be calibrated at the factory before shipping it is suggested that you recalibrate the probe before using it to set any tools. Figure 7.1.1 Manual Tool Setting Arm and Probe 8.1.2 Automatic Tool Setting Arm and Probe An automatic tool setting arm and probe is fitted to the TC820LTYsi machine as standard. The tool setting arm is automatically deployed by the button on the MCP or with M-Codes. The automatic probe can be used in the jog environment to set tool data or in auto to check for tools throughout a part program or production run. The position of the tool setting arm and probe are stored in the machine and used to set the position of the tools cutting edge with reference to the work piece and machine coordinate system. If the tool setting arm or probe is ever crashed, it will require recalibration. Although your probe will be calibrated at the factory before shipping it is suggested that you recalibrate the probe before using it to set any tools. Figure 7.1.2 Automatic Tool Setting Arm and Probe 8.2 Tool Probe Calibration It is recommended you calibrate your tool setting arm and probe as soon as you receive your TCsi machine from the factory. The probe will also need to be recalibrated if a collision occurs. 8.2.1 Setting Up the Tool Table In order to calibrate all 4 sides of the tool setting probe 3 different tools will need to be used. To calibrate the tool setting probe, set up: • An OD roughing tool in position #1 on the turret. • An ID roughing tool in position #3 on the turret. • A cutoff tool in position #5 on the turret Populate your tool table with the information for the 3 tools installed on the turret following the instructions below, see figure 7.2a for a sample of the completed Tool list:

SOP-TC820LTYsi-317 自动刀具设定臂和探头校准	
部件/系统 component	automatic tool setter probe
安全等级 safety_level	critical
前置条件 preconditions	• 刀具表和 G500 已准备 • 门互锁正常
作业步骤 steps	1. 选择 Meas. Tool → Calibrate。 2. 用 MCP 部署自动刀具设定臂。 3. 将适用刀具移至所选探头边缘约 0.5 in。 4. Cycle Start 触碰并更新数据。 5. 对四个探头面重复。
安全要求 safety_warnings	• 自动探头只能门关闭时部署 • 起始距离约 0.5 in 且无干涉
验收标准 acceptance_criteria	• 四面数据更新且无碰撞/报警。
关键词 keywords	四面校准 • 0.5 in • 探头
来源 page / page_end / section_code	手册 LTY-SOP • P125–P126
原文摘录 source_excerpt	8.2.3 Calibrate the Tool Setting Arm and Probe To calibrate the tool setting arm and probe press [M] from the left VSK. Make sure any adjacent tools will not hit the probe or chuck during the calibration process. Press “Meas. Tool” from the HSK and “Calibrate” from the right VSK. Press one of the buttons on the right VSK for the direction to be calibrated shown in figure 7.2.3a. Figure 7.2.3a Probe Calibration Screen • On the TC820si for the manual tool setter there is a release knob inside of the machine envelope just under the chuck. Pull the knob as shown in figure 7.2.3b. Swing the probe arm into position as shown in figure 7.2.3c. There is a light on the probe to show it is active, this only works when the door is closed. Figure 7.2.3b Release Knob Figure 7.2.3c Probe in Position

SOP-TC820LTYsi-318 JOG 模式探头测量刀具	
部件/系统 component	automatic tool setter probe
安全等级 safety_level	high
前置条件 preconditions	• 探头已校准 • 相邻刀具无干涉
作业步骤 steps	1. 选择 Meas. Tool → Automatic。 2. 选择方向并查看提示。 3. 用 MPG 移到探头前约 1/2 in。 4. Cycle Start 完成接触并重复另一轴。 5. 确认 Tool data 更新。
安全要求 safety_warnings	• MPG 低速靠近并确认路径 • 异常立即停止
验收标准 acceptance_criteria	• 目标刀具 X/Z 数据更新。
关键词 keywords	JOG • MPG • Tool data
来源 page / page_end / section_code	手册 LTY-SOP • P126–P127
原文摘录 source_excerpt	• On the TC820LTYsi for the automatic tool setter the tool setting arm is automatically deployed by pressing the button on the MCP, figure 7.2.3d, or through the use of M-Codes. The arm will swing into position as shown in figure 7.2.3e. The probe can only be deployed when the door is closed. Figure 7.2.3d Probe Deploy Button Figure 7.2.3e Probe in Position All 4 sides of the tool setter will need to be calibrated, move the tool to within 0.5” of the edge you selected on the calibration screen on the machine control, figure 7.2.3a, and press cycle start. The tool should move and touch the probe updating the probe data. Repeat this process for all 4 sides of the probe using the appropriate tool that can reach and touch off the desired side. 8.3 Measuring Tools Using the Tool Setting Arm and Probe in Jog Press [M] from the left VSK. Make sure any adjacent tools will not hit the probe or chuck during the measurement cycle. Press “Meas. Tool” from the HSK and “Automatic” from the right VSK. Press one of the buttons on the right VSK for the direction to be measured. The screen will display which of the four sides of the probe need to be touched by the tool as shown in figure 7.3a. Use MPG to move the active tool about ½” away from the correct probe. Press cycle start. Figure 7.3a First Position

SOP-TC820LTYsi-319 AUTO 模式探头测量刀具	
部件/系统 component	automatic tool setter probe
安全等级 safety_level	high
前置条件 preconditions	• 探头已校准 • 零件程序已验证
作业步骤 steps	1. 在 Auto 调用 Siemens 测量循环。 2. 核对刀塔布置和相邻工具间隙。 3. 单件低风险试运行。 4. 确认结果写入正确刀具补偿。
安全要求 safety_warnings	• 嵌入生产程序前验证所有干涉 • 报警时停止程序检查反馈
验收标准 acceptance_criteria	• Auto 测量循环无碰撞和报警。
关键词 keywords	AUTO • 测量循环 • 刀具补偿
来源 page / page_end / section_code	手册 LTY-SOP • P127–P127
原文摘录 source_excerpt	Repeat this process for the other axis displayed on the right VSK as shown in figure 7.3b. Figure 7.3b Second Position This process will fill in the dimensions in the “Tool data” chart in the Work Area. Repeat this same process using whichever tool needs to be measured. 8.4 Measuring Tools Using the Tool Setting Arm and Probe in Auto The TC820LTY automatic tool setting arm and probe with the Siemens measuring cycle software can be used in the Auto environment and embedded in a part program and production run. Before the probe measuring cycle can be embedded in a program the tool setting arm and probe needs to be calibrated. Pay attention to the tooling set up on the turret so any tool being measured will not have other tools interfere and hit the tool setting arm or the chuck. 8.5 Tool Setting Probe Troubleshooting The probe is designed to turn on when the tool setting arm is deployed. When the probe signal is not working for the TC820si manual tool setting arm and probe: • Check the door is closed and the interlock is engaged. The door open message on the control should not be showing. • Check the connecting cable in the X222 area of the TC820si electrical cabinet and make sure the red wire is connected properly to X222-216-1 and the white wire is connected properly to X132.13- 8. • There are limit switches that show when the probe is deployed and parked. Refer to the TRAK I/O Diagnostic screen and make sure all the inputs and outputs for the probe and tool setting arm limit switches are working as expected. When the probe signal is not working for the TC820LTYsi automatic tool setting arm and probe: • Check the door is closed and the interlock is engaged. The door open message on the control should not be showing.

SOP-TC820LTYsi-320 自动探头信号故障检查	
部件/系统 component	automatic tool setter probe
安全等级 safety_level	high
前置条件 preconditions	• 探头信号异常且机床停止相关动作
作业步骤 steps	1. 确认门关闭且互锁有效。 2. 检查 TSI 3 到 X111、X222、TB4 的电缆。 3. 在 TRAK I/O Diagnostic 观察限位输入输出。 4. 参考 Renishaw 手册并保存报警/I/O证据。
安全要求 safety_warnings	• 不得带电插拔电柜连接器 • 门互锁异常不得旁路运行
验收标准 acceptance_criteria	• 部署、停车和触发反馈与实际状态一致。
关键词 keywords	TSI 3 • X111 • X222 • TB4
来源 page / page_end / section_code	手册 LTY-SOP • P127–P128
原文摘录 source_excerpt	Repeat this process for the other axis displayed on the right VSK as shown in figure 7.3b. Figure 7.3b Second Position This process will fill in the dimensions in the “Tool data” chart in the Work Area. Repeat this same process using whichever tool needs to be measured. 8.4 Measuring Tools Using the Tool Setting Arm and Probe in Auto The TC820LTY automatic tool setting arm and probe with the Siemens measuring cycle software can be used in the Auto environment and embedded in a part program and production run. Before the probe measuring cycle can be embedded in a program the tool setting arm and probe needs to be calibrated. Pay attention to the tooling set up on the turret so any tool being measured will not have other tools interfere and hit the tool setting arm or the chuck. 8.5 Tool Setting Probe Troubleshooting The probe is designed to turn on when the tool setting arm is deployed. When the probe signal is not working for the TC820si manual tool setting arm and probe: • Check the door is closed and the interlock is engaged. The door open message on the control should not be showing. • Check the connecting cable in the X222 area of the TC820si electrical cabinet and make sure the red wire is connected properly to X222-216-1 and the white wire is connected properly to X132.13- 8. • There are limit switches that show when the probe is deployed and parked. Refer to the TRAK I/O Diagnostic screen and make sure all the inputs and outputs for the probe and tool setting arm limit switches are working as expected. When the probe signal is not working for the TC820LTYsi automatic tool setting arm and probe: • Check the door is closed and the interlock is engaged. The door open message on the control should not be showing.

SOP-TC820LTYsi-321 电柜热交换器过滤器清洁	
部件/系统 component	electrical cabinet cooling
安全等级 safety_level	high
前置条件 preconditions	• 机床停机并断电 • 过滤器可安全取出
作业步骤 steps	1. 抬底部卡舌取出过滤器。 2. 用空气吹净或水洗。 3. 清洁风扇格栅散热片。 4. 干燥后滑回槽内固定。
安全要求 safety_warnings	• 过滤器干燥后再装回 • 不得带电接触电柜内部
验收标准 acceptance_criteria	• 过滤器和冷却通道无堵塞。
关键词 keywords	电柜 • 热交换器 • 过滤器
来源 page / page_end / section_code	手册 LTY-SOP • P84–P84
原文摘录 source_excerpt	5.0 Maintenance and Replacement 5.1 Maintenance 5.1.1 How to Clean the Touchscreen & Machine Control Panel (MCP) The pendant is designed for low maintenance operation. Nevertheless, you must regularly clean the pendant. WARNING!!! 1. Turn power off when cleaning the pendant. 2. Do not clean the device using compressed air or steam jets because they can damage it. Clean only the front panel with a soft cloth moistened with water or mild cleaning agent. Spray the cleaning agent or water on to a cleaning cloth. Do not spray anything directly on the front panel. Never use caustic solvents or abrasive cleaners. Use dry cloth to clean the backside of the pendant. 5.1.2 How to Clean the Electrical Cabinet Heat Exchanger Air Filter 1. Lift tab at the bottom of the filter which lifts the filter out of the slot in front of the fan, see figure 4.1.2. 2. Clean filter either by blowing with air or rinsing with water. 3. Clean fins on the grate of the fan. 4. Dry filter if needed and replace into the slot and slide down to secure. Figure 4.1.2 Heat Exchanger Filter

SOP-TC820LTYsi-322 液压泵过滤器更换	
部件/系统 component	hydraulic system
安全等级 safety_level	critical
前置条件 preconditions	• 系统断电泄压 • 准备起吊设备和新过滤器
作业步骤 steps	1. 拆六个螺钉和两个吊环螺栓。 2. 用起吊设备分离并提升上盖。 3. 更换过滤器。 4. 复位上盖并固定油箱。
安全要求 safety_warnings	• 油箱上盖是重物必须起吊 • 防止油液喷出和污染
验收标准 acceptance_criteria	• 过滤器已换、密封正常、无泄漏。
关键词 keywords	液压泵 • 过滤器 • 起吊设备
来源 page / page_end / section_code	手册 LTY-SOP • P85–P85
原文摘录 source_excerpt	5.1.3 How to Replace Hydraulic Pump Filter WARNING!!! Heavy Object, us proper lifting equipment. 1. Remove the six screws and two eye bolts between the top cover and the tank, see figure 4.1.3. 2. Use proper lifting equipment to separate the top cover from the tank. 3. Lift the top cover high enough to replace the filter. 4. Replace top cover and secure to tank. Figure 4.1.3 Hydraulic Pump 5.2 Maintenance Intervals 5.2.1 Daily Maintenance • Remove majority of chips and debris from around the ways, way covers, chuck, turret, tool holders and other moving parts. • Empty chips from screen sitting above the coolant tank. • Check hydraulic oil levels. If the oil level is low check for leaks in the hydraulic system. If needed fill with ISO VG32/SAE 10W. • Check the main hydraulic pressure is set to 700psi. • Check the lubrication pump oil level and make sure it is always above the minimum line. If needed fill with ISO VG68/SAE 20W. • Check to make sure there are no obvious oil leaks to the lubrication system, and it is getting to all the slideways and moving parts. • Check lubrication pressure is set to 70psi when the pump is running. • Check to make sure the clamping mechanism is clean and in proper working order. Grease the chuck twice daily. • Check coolant level and add if low.

SOP-TC820LTYsi-323 各轴和尾座镶条调整	
部件/系统 component	gibs
安全等级 safety_level	high
前置条件 preconditions	• 轴停止且区域清洁 • 确认对应轴和镶条类型
作业步骤 steps	1. 检查镶条平直、无毛刺且油道畅通。 2. TC820LTYsi X/Y/Z 从后盖、B 轴从尾座防护板进入。 3. 用扭矩螺丝刀设为 1.2 Nm/10.62 lbf.in。 4. 按单面/双面镶条步骤往复、复紧和锁定。 5. 复装后检查运动、润滑和精度。
安全要求 safety_warnings	• 错误调整会损失精度 • 禁止用垫片替代，避免气流吹屑入滑道
验收标准 acceptance_criteria	• 运动平稳，调整扭矩和精度符合要求。
关键词 keywords	X轴 • Y轴 • Z轴 • B轴 • 1.2 Nm
来源 page / page_end / section_code	手册 LTY-SOP • P86–P92
原文摘录 source_excerpt	5.2.2 Monthly Maintenance • Clean air filters on the electrical cabinet heat exchanger and replace if needed. • Remove covers and inspect ways, clean areas of chips and debris. • Every 2 months remove and drain the coolant tank, clean inside and fill with new clean coolant. 5.2.3 Half Yearly Maintenance • Check all switches and interlocks are working correctly. • Check the level of the machine and re-level as necessary. • Check the spindle motor belt tension and adjust as necessary. • Check the backlash on each axis and adjust as necessary. • Check gibs on each axis and tailstock and adjust as necessary. 5.2.4 Yearly Maintenance • Remove the hydraulic tank. Drain oil and clean inside. Clean filter and fill with new clean hydraulic oil (ISO VG32/SAE 10W). • Remove all covers, vacuum chips and debris that may have built up, and wipe down machine from top to bottom. • Inspect machine for any unusual wear and play. • Check cables, coolant hoses and hydraulic lines for any abrasion or cuts. 5.3 Mechanical Adjustments 5.3.1 Gib Adjustments WARNING!!! Ensure to adjust gibs frequently and properly. Otherwise, this might result in loss of machine accuracy. The gibs are vital to the performance of your TCsi Machine. Gibs should be: • flat • free of twist • free of burrs • free of blockages in the oil passages and channels Defective or scarred gibs must be replaced. Shimming of gibs will not yield acceptable results. It is good machining practice to avoid the use of shop air to clean the chips off a machine. This risks blowing chips into the sliding way surfaces and compromising the performance of the machine. Because of long-term friction between the bed and carriage, wear may occur. Proper gib adjustment is necessary after the machine has been operated for a long time. Adjust the gib by turning the gib adjustment screw (2) using torque screwdriver see figure 4.3.1. Set torque to 1.2Nm (0.9 ft lbs).

SOP-TC820LTYsi-324 主轴皮带检查和重新张紧	
部件/系统 component	spindle belt
安全等级 safety_level	critical
前置条件 preconditions	• 确认需检查/调整皮带 • 电源已断开并防止启动
作业步骤 steps	1. 松开电机板六个螺钉并调张紧器。 2. TC820LTYsi 仪表输入 150 g/m、11 mm/#R、span 766 mm。 3. 测量目标约 120 lbf, 允许 105–145 lbf, 新带约 145 lbf。 4. 确认后六螺钉紧固 100 ft-lbs。
安全要求 safety_warnings	• 无法确保张力时不要调整 • 新皮带 48 小时后复查
验收标准 acceptance_criteria	• 张力、对中和扭矩符合要求，运行无异常噪声。
关键词 keywords	766 mm • 120 lbf • 105-145 lbf • 100 ft-lbs
来源 page / page_end / section_code	手册 LTY-SOP • P94–P96
原文摘录 source_excerpt	5.4.3 Check Spindle Belt WARNING!!! Ensure to have a proper tension value for the spindle transmission belt. If you cannot ensure the proper tension value, please do not adjust the belt tension. After the machine has been operated for a long period, the spindle drive-timing belt may gradually become loose. Check the main drive belt tension frequently. 5.4.4 Re-tension Spindle Belt Follow steps below to adjust the belt tension reference figure 4.4.2a. Two tool choices are explained to test the tension: 1. Make sure the power source has been disconnected before adjusting the timing belt tension. 2. Loosen the 6 fastening screws (1) on the spindle motor mounting plate. Figure 4.4.2a Spindle Belt Orientation 3. Set belt tension by adjusting the belt tension motor bracket reference figure 4.4.2b inset photo. Use the tensioner block to either increase tension on the belt or decrease tension. When the tension on the belt looks good, snug the 6 fastening screws.

SOP-TC820LTYsi-325 X/Z 轴反向间隙参数检查	
部件/系统 component	axis backlash
安全等级 safety_level	high
前置条件 preconditions	• 确认重复性/定位精度需检查 • 具备制造商级参数权限
作业步骤 steps	1. 进入 Setup → Mach data → Axis MD。 2. Search 参数 32450。 3. Select axis 查看 AX1/AX2。 4. 记录参数和测量结果，授权后才可调整。
安全要求 safety_warnings	• 32450[0] 是单机值，禁止套用其他机床 • Manufacturer level password 才可编辑
验收标准 acceptance_criteria	• 参数与单机实测和维修记录一致。
关键词 keywords	32450 • AX1 • AX2
来源 page / page_end / section_code	手册 LTY-SOP • P97–P97
原文摘录 source_excerpt	5.4.5 Backlash Constants Backlash settings for the TCsi machines are set by the factory during the build for the X & Z-Axis. These values are unique for each machine. The backlash values for the two axes, AX1 (X-Axis) and AX2 (Z-Axis) are stored in the control under Axis MD parameter 32450[0]. To find the parameter, press “Setup” , make sure you are on page 1 according to the far-right corner of HSK, press “Mach data” then “Axis MD” . From the right VSK press “Search” and enter 32450 into the field. The backlash values may need to be adjusted over time to account for wear in the drive train system, see figure 4.4.3a. Figure 4.4.3a Axis MD Using the right VSK press “Select axis” and you will see the axis list appear. You need “Manufacturer” level password to be able to edit the fields, see figure 4.4.3b. Figure 4.4.3b Select Axis

SOP-TC820LTYsi-326 泵式油水分离器排油	
部件/系统 component	pump type oil skimmer
安全等级 safety_level	high
前置条件 preconditions	• 已安装泵式撇油器 • 准备废油容器
作业步骤 steps	1. 检查分离油液位。 2. 打开阀门排入收集容器。 3. 关闭阀门并按环保规定处置。 4. 检查回流路径和溢漏。
安全要求 safety_warnings	• 防止分离油溢回冷却液箱 • 拆泵/皮带前先急停断电并装防护
验收标准 acceptance_criteria	• 分离油未溢回冷却液箱，阀门和回流正常。
关键词 keywords	撇油器 • 分离器 • 排油
来源 page / page_end / section_code	手册 LTY-SOP • P130–P131
原文摘录 source_excerpt	9.1.2 Pump Type Oil Skimmer The pump type oil skimmer on the TC820LTYsi machines is positioned adjacent to the coolant pump. The oil skimmer is wired directly to the electrical cabinet and is controlled using a button on the MCP or through the use of M-Codes. As the pump of the oil skimmer runs the top layer of the coolant is removed with the layer of oil and dumped in the oil separator. The oil and coolant collected by the oil skimmer is then separated with the coolant being returned back into the coolant tank and the oil passed into a section of the oil separator where it can be removed. The oil skimmer helps keep your coolant free of oil, extending the life of the coolant and reducing coolant maintenance. Figure 8.1.2 shows an overview of the pump type oil skimmer on the TC820LTYsi machine. Figure 8.1.2 Pump Type Oil Skimmer 9.1.3 Maintenance 9.1.3.1 Emptying the Skimmed Oil Both oil skimmer options on the TCsi machines come with a collection container. • On the TC820si machines the skimmed oil will run off the oil separator into the collection container. There is a bucket beneath it to catch any excess oil, but it is important to empty the skimmed oil before it spills over the top. Simply remove the funnel and pick up the container using the handle and empty the oil where it can be disposed. Any oil skimmed while the tank is removed will be captured in the bucket below the tank. • On the TC820LTY machines the skimmed oil will be held in the oil separator. There is a valve that can be opened to empty the skimmed oil from the oil separator into the collection container. It is important to empty the skimmed oil from the oil separator into the collection container before it runs over the top where it will then return to the coolant tank. Simply open the valve to empty

SOP-TC820LTYsi-327 接料器清洁和功能检查	
部件/系统 component	part catcher
安全等级 safety_level	high
前置条件 preconditions	生产结束且接料器处于安全状态
作业步骤 steps	- 清除接料器本体和滑道切屑。 - 清空储料桶，防止拥堵。 - 清除桶内和筛网下方切屑。 - 用 MCP 测试升起和闲置位置及反馈。
安全要求 safety_warnings	- 停止动作后清洁 - 堵塞回流路径会导致冷却液溢出
验收标准 acceptance_criteria	升降无迟滞、位置反馈正确、回流畅通。
关键词 keywords	接料器 • 切屑 • 冷却液回流
来源 page / page_end / section_code	手册 LTY-SOP • P132–P133
原文摘录 source_excerpt	5. Loosen or remove as needed the two screws holding the pulley bracket see figure 8.1.2.2c in order to compress the spring and remove the belt. Figure 8.1.2.2a Mounting Bolts Figure 8.1.2.2b Cover Screws Figure 8.1.2.2c Pulley 5. Replace the belt and reverse the above procedure to replace the oil skimmer. 9.2 Part Catcher 9.2.1 Programable Part Catcher The TRAK part catcher on the TCsi machines is operated using the button on the MCP or through the use M-Codes embedded in a program. Driven by the hydraulic system, the part catcher can be deployed to catch finished parts as they are cut off and retracted to be clear during machining operations. Utilizing the part catcher can aid with unattended machining as the finished part runs down the shoot to be stored in a built-in bucket, see figure 8.2.1.

SOP-TC820LTYsi-328 高压冷却泵更换	
部件/系统 component	high-pressure coolant
安全等级 safety_level	critical
前置条件 preconditions	• 确认高压泵故障 • 急停/断电并准备接液和起吊设备
作业步骤 steps	1. MCP 关闭流体并关闭电源。 2. 排液至过滤器罐清空。 3. 断开泵电缆和传感器线并记录方向。 4. 拆过滤器、螺钉并吊出旧泵。 5. 装入新泵，复接管线和电缆。 6. 启动后核对旋向，至少无故障运行 10 秒并查漏。
安全要求 safety_warnings	• 约 220 psi，先关闭流体和电源并排液 • 泵总成是重物必须起吊
验收标准 acceptance_criteria	• 约 220 psi、旋向正确、10 秒无故障且无泄漏。
关键词 keywords	220 psi • 起吊 • 10 秒
来源 page / page_end / section_code	手册 LTY-SOP • P134–P135
原文摘录 source_excerpt	9.3.1 High Pressure Coolant The TRAK high pressure coolant system, see figure 8.3.1, delivers coolant at 220 psi directly through the turret and cutting tool effectively dissipating chips and heat during machining. The increased pressure will allow for heavier cuts, higher feed rates, deep-hole drilling, extended tool life and better surface finishes. This produces reduced cycle times, increase throughput, and higher-quality parts. The high-pressure coolant pump is fitted with a filter and pressure sensor, this helps keep the filter clean and at enough pressure to ensure optimal cutting conditions. Figure 8.3.1 High Pressure Coolant 9.3.2 High Pressure Coolant Pump The high-pressure coolant system uses a larger pump delivering coolant at 220 psi. If the pump fails it will need to be replaced. Warning!!! Heavy object to be installed, proper safety precautions are needed. To replace a faulty pump: 1. Turn off fluid flow using button on the MCP and turn off power to the TCsi machine. 2. Drain enough coolant to clear the filter canister. 3. Disconnect the cable going from the electric cabinet to coolant pump and the wires at the pressure sensor, see figure 8.3.2. Note orientation of the sensor wires.

SOP-TC820LTYsi-329 高压冷却压力传感器更换	
部件/系统 component	high-pressure coolant
安全等级 safety_level	critical
前置条件 preconditions	• 流体已关闭且电源隔离 • 记录原线方向
作业步骤 steps	1. 关闭流体和电源。 2. 移开滑盖并拆 captive screw。 3. 断开两个 ferrule，记录方向并拆传感器。 4. 安装新传感器。 5. 设为 85 psi，启动冷却至少 10 秒查漏。
安全要求 safety_warnings	• 未释放压力不得拆卸 • 更换后查漏
验收标准 acceptance_criteria	• 85 psi、反馈正常且无泄漏。
关键词 keywords	85 psi • 压力传感器 • 高压冷却
来源 page / page_end / section_code	手册 LTY-SOP • P135–P135
原文摘录 source_excerpt	Figure 8.3.2 High Pressure Coolant Wiring 4. Prepare to collect the coolant and then disconnect the coolant hose from the pressure sensor. 5. Remove the filter with pressure sensor from the coolant pump. 6. Remove four SHCS screws that bolt the motor to the sheet metal base. 7. Using a proper lifting device, slowly lift the pump assembly from the coolant tank allowing it to drain thoroughly. Set down in a safe manner. 8. Using the lifting device, carefully guide the new pump assembly into position and reconnect the four SHCS. 9. Reconnect the filter and pressure sensor assembly to the pump outlet. The sensor should not need recalibrating. 10. Reconnect coolant hose using Teflon tape. 11. Connect wires to the pressure sensor. Plug in the pump electric cable to electric cabinet. 12. Power up the TCsi machine and turn on the coolant using the button on the MCP. Verify the motor is turning in the correct direction with the arrows on top of the pump and make sure the coolant comes on and operates for at least 10 seconds without faulting. 9.3.3 High Pressure Coolant Pressure Sensor The coolant pressure sensor checks for proper coolant pressure through the system. The coolant pressure sensor is designed to warn if there is insufficient pressure in the system, it monitors the coolant pressure in case of coolant runout or equipment failure. Under these conditions there is a warning message displayed to give the operator a chance to correct the problem. If this warning message is triggered, the sensor halts the current program and stops operation in the machine. 9.3.3.1 Pressure sensor replacement: 1. Turn off fluid flow using the button on the MCP and turn off power to the TCsi machine. 2. Loosen the slide cap and unscrew the captive screw at the top of the sensor. Now you can slide the cap out of the way, see figure 8.3.2. 3. Disconnect two small electric ferrules on the sensor and remove the wires then loosen the sensor with a wrench. Note the wire orientation. Replace the old sensor with the new one. 4. Reverse above steps to reconnect the sensor. 5. Set the pressure value of the pressure sensor to 85psi (6 kg/cm2). 6. Power up the TCsi machine and turn on the coolant using the button on the MCP. Make sure the coolant comes on and operates for at least 10 seconds without faulting. 9.3.4 High Pressure Coolant Filter Element The filter element is an important safeguard to keep contaminants out of the coolant system. 9.3.4.1 Filter Element Replacement: The filter for the high-pressure coolant system should be cleaned every time the pressure sensor triggers or you replace your coolant:

SOP-TC820LTYsi-330 高压冷却过滤元件更换	
部件/系统 component	high-pressure coolant
安全等级 safety_level	high
前置条件 preconditions	• 压力传感器触发或更换冷却液后执行 • 系统断电隔离
作业步骤 steps	1. 关闭系统并拆压力传感器。 2. 拆滤器顶部四个螺钉并托住壳体。 3. 取出滤芯，检查并润滑 O-ring。 4. 装新滤芯、滤罐并交叉紧固。 5. 用 Teflon tape 复装传感器，通电查漏。
安全要求 safety_warnings	• 保护 O-ring 防止泄漏 • 通电前确认滤罐和压力表牢固
验收标准 acceptance_criteria	• O-ring、滤芯、密封和泄漏检查通过。
关键词 keywords	O-ring • 滤芯 • Teflon tape
来源 page / page_end / section_code	手册 LTY-SOP • P135–P136
原文摘录 source_excerpt	Figure 8.3.2 High Pressure Coolant Wiring 4. Prepare to collect the coolant and then disconnect the coolant hose from the pressure sensor. 5. Remove the filter with pressure sensor from the coolant pump. 6. Remove four SHCS screws that bolt the motor to the sheet metal base. 7. Using a proper lifting device, slowly lift the pump assembly from the coolant tank allowing it to drain thoroughly. Set down in a safe manner. 8. Using the lifting device, carefully guide the new pump assembly into position and reconnect the four SHCS. 9. Reconnect the filter and pressure sensor assembly to the pump outlet. The sensor should not need recalibrating. 10. Reconnect coolant hose using Teflon tape. 11. Connect wires to the pressure sensor. Plug in the pump electric cable to electric cabinet. 12. Power up the TCsi machine and turn on the coolant using the button on the MCP. Verify the motor is turning in the correct direction with the arrows on top of the pump and make sure the coolant comes on and operates for at least 10 seconds without faulting. 9.3.3 High Pressure Coolant Pressure Sensor The coolant pressure sensor checks for proper coolant pressure through the system. The coolant pressure sensor is designed to warn if there is insufficient pressure in the system, it monitors the coolant pressure in case of coolant runout or equipment failure. Under these conditions there is a warning message displayed to give the operator a chance to correct the problem. If this warning message is triggered, the sensor halts the current program and stops operation in the machine. 9.3.3.1 Pressure sensor replacement: 1. Turn off fluid flow using the button on the MCP and turn off power to the TCsi machine. 2. Loosen the slide cap and unscrew the captive screw at the top of the sensor. Now you can slide the cap out of the way, see figure 8.3.2. 3. Disconnect two small electric ferrules on the sensor and remove the wires then loosen the sensor with a wrench. Note the wire orientation. Replace the old sensor with the new one. 4. Reverse above steps to reconnect the sensor. 5. Set the pressure value of the pressure sensor to 85psi (6 kg/cm2). 6. Power up the TCsi machine and turn on the coolant using the button on the MCP. Make sure the coolant comes on and operates for at least 10 seconds without faulting. 9.3.4 High Pressure Coolant Filter Element The filter element is an important safeguard to keep contaminants out of the coolant system. 9.3.4.1 Filter Element Replacement: The filter for the high-pressure coolant system should be cleaned every time the pressure sensor triggers or you replace your coolant:

SOP-TC820LTYsi-331 LNS 棒料送料器安全操作和故障检查

部件/系统 component	barfeeder
安全等级 safety_level	critical
前置条件 preconditions	• 已安装 Barfeed Ready • 操作员完成 LNS/机床安全培训
作业步骤 steps	1. 检查主电源、急停、停止灯和磁性安全开关。 2. 检查主入口盖、保护盖和锁扣。 3. 核对棒料 6–80 mm、300–1600 mm。 4. 排查 I6.0/I6.2/I6.6 和 Easy Extend 配置。 5. 无法恢复时升级受训维修。
安全要求 safety_warnings	• 维修仅由 TRAK/LNS 受训人员执行 • 禁止手伸入 magazine、保护盖和送料机构 • 急停和磁性安全开关必须有效
验收标准 acceptance_criteria	• 安全开关、反馈、选项配置和试运行均正常。
关键词 keywords	LNS • 6-80 mm • 300-1600 mm • I6.0 • I6.2 • I6.6
来源 page / page_end / section_code	手册 LTY-SOP • P136–P140
原文摘录 source_excerpt	1. Turn off system power. 2. Remove pressure sensor, see section 8.3.3.1 Carefully remove gauge with a wrench, see figure 8.3.4.1a. 3. Remove the four socket head cap screws from the top of the filter body, be ready to catch the element housing and don't lose the O- ring. 4. Slide off the filter element from the housing. 5. Check the canister O-ring and replace if needed. Figure 8.3.4.1a Gauge 6. Apply lubricant to the O-ring with petroleum jelly or light oil and place on filter head. 7. Slide the new filter on the housing, install canister and make sure O- ring is in its groove, see figure 8.3.4.1b. 8. Cross tighten the screws. 9. Replace the gauge and pressure sensor using Teflon tape. 10. Confirm there are no leaks after powering the system. Figure 8.3.4.1b O-ring 9.4 LNS Quick Load Servo 80 S2 Barfeed and Barfeed Ready 9.4.1 LNS Barfeed Operation The LNS Quick Load Servo 80 S2 Bar Feeder allows for continuous machining. With quick changeovers from one part run to another and a library of up to 500-part programs, this servo driven bar feed will drastically increase productivity. Equipped with an automatic diameter changing system, no manual adjustment is required reducing changeover time to a minimum. The LNS servo motor control technology guarantees accurate bar feeding into the machine and prevents hitting the turret with its soft touch positioning.

SOP-TC820LTYsi-332 三爪液压卡盘拆装	
部件/系统 component	hydraulic chuck
安全等级 safety_level	critical
前置条件 preconditions	- 卡盘需要拆装且已急停 - 准备吊装设备和吊环
作业步骤 steps	- 松开卡盘并急停，拆除卡爪。 - 用吊环起吊卡盘，拆下 draw nut。 - 清洁安装面、检查高点并涂防咬合剂。 - 用吊装设备对齐 drawtube，手紧六个螺钉。 - 按 53.5 ft-lbs 紧固并完成复装。
安全要求 safety_warnings	- 卡盘是重物必须吊装 - 安装后六螺钉 53.5 ft-lbs
验收标准 acceptance_criteria	安装面、扭矩、夹紧和运行测试通过。
关键词 keywords	卡盘 • 53.5 ft-lbs • drawtube
来源 page / page_end / section_code	手册 LTY-SOP • P66–P66
原文摘录 source_excerpt	Installation Checklist - Press the oil skim button to make sure the skimmer starts to skim oil. - Verify the oil skimmer pump is rotating in the clockwise direction. - Press the oil skim button again to make sure the skimmer stops. - Attach the way lube runoff hose to the oil separator box. Part Catcher – See part catcher section 9.2 of the Service Manual. - Test functionality using buttons on MCP. - Press the parts button to make sure the part catcher moves to its raised position, press button again to make sure it returns to its idle position. High Pressure Coolant – See high pressure coolant section 8.3. - Wire up the high-pressure coolant pressure sensor. - Test filter pressure sensor: - With the pump turned on and the pressure sensor unplugged check the IO screen in TRAK diagnostic to see coolant pressure switch is false. - Calibrate the coolant pressure sensor: - Remove any tool from the turret prior to calibration. - Using a screwdriver slowly tighten the screw on the top center of the sensor approximately 1/8th of a turn per second. - When the machine faults out, loosen the screw ¼ turn, and clear the fault on the pendent. Run the high-pressure coolant pump for 10 minutes and check for leaks. 3.15 Jaw Hydraulic Chuck Mounting 3.15.1 Instructions to remove the 3 Jaw Chuck: - Unclamp the chuck and press the E-Stop in center of travel. - Remove 2 of the 3 jaws from the chuck. - Using the remaining jaw to hold the chuck in place loosen all 6 screws holding the chuck to the spindle face. - Remove the last jaw from the chuck. - Remove chip cover from the center of the chuck. - Attach lifting eye bolt and hoist the chuck to take pressure off the drawtube and nut. - Remove the 6 screws holding the chuck. - Unscrew draw nut by hand, you may need to adjust the hoist until the nut rotates by hand. 3.15.2 Instructions to reinstall the 3 Jaw Chuck: - Clean off all the mounting surfaces and threads with WD40 and make sure they are free from any chips or swarf. - Use a fine grit deburring stone to check for dinks or high spots on the mounting faces. - Apply anti seize to the threads and mounting screws. - Set the draw tube to the mid travel position and E-stop the machine. - Using the eyebolt on the chuck, hoist it into position and align with the draw nut with the drawtube. - Screw the draw nut by hand until it is approximately ¾ engaged on the drawtube thread. - Align the chuck body holes to the holes on the spindle face and hand tighten the 6 screws. - Remove the hoist so the chuck is supporting itself. - Attach one jaw to the chuck and use it to torque the 6 mounting screws to 53.5 ft lbs. - Remove the jaw again from the chuck and thread the draw nut until it bottoms against the bottom of the drawtube thread.

ql80s2_sop_safe_service_entry 检修前安全隔离与复机前检查	
触发条件 trigger	任何拆盖、调整、清洁、维修或进入运动机构区域前
前置条件 preconditions	- 由受过培训的人员执行；电气故障由合格电工处理 - 确认工作区域照明良好、地面无油污和障碍
作业步骤 steps.action / steps.safety	1. 停止棒料机和车床，结束自动循环。 安全：确认棒料和夹紧装置处于稳定状态。 2. 关闭棒料机主电源；维修涉及运动或气动机构时关闭气源并释放压力。 安全：不得在带电或带压状态下拆除防护盖。 3. 拆盖前确认棒料机已与车床脱开，并隔离可能的车床启动信号。 安全：防止跨设备误启动。 4. 完成维修后安装全部防护件，清理工具和异物，检查电气部件未接触水/油。 安全：未恢复安全防护不得试运行。 5. 按对应 SOP 进行低速/手动验证，再恢复自动循环。 安全：先验证安全联锁和急停，再验证送料。
安全要求 ppe_and_safety	- 关闭棒料机电源并断开电源插头 - 关闭并释放气源/确认无气动压力 - 拆盖前将棒料机与车床脱开 - 不得把手伸入运动中的棒料机 - 不得踩踏皮带 - 不得拆除、短接或旁路安全开关/断路器 - 维修完成后必须恢复安全防护件
工具与物料 tools_and_materials	- 设备断电/气源隔离设施 - 适用的检修工具 - LNS 推荐或提供的备件
验收标准 acceptance_criteria	- 防护件完整并锁好 - 急停和安全开关功能正常 - 无遗留工具、油污或异物 - 手动验证无异常阻力/刮擦后才允许自动运行
回滚与升级 rollback_or_escalation	电气故障、保护回路异常或无法确认安全状态时停止操作并联系 LNS 或当地代表。
来源 evidence.page / evidence.section	手册 P9 • SAFETY INSTRUCTIONS • 置信度 high
原文摘录 evidence.evidence_text	Always switch off before service; uncouple before removing covers; no servicing while powered; do not bridge safety switches

ql80s2_sop_alignment_anchor 棒料机与车床对中及锚固	
触发条件 trigger	首次安装、设备移动后、报警45持续出现或发现棒料在主轴内刮擦/卡滞
前置条件 preconditions	- 棒料机断电 - 无气动压力 - 车床稳定且已找平 - 必要时确认棒料机与主轴后端间距不超过 200 mm，超出时准备延长件
作业步骤 steps.action / steps.safety	- 将棒料机尽量靠近车床后部放置，用水平仪检查 X/Y 方向；在直线皮带导轨和退让导轨处测量。 - 确认中心顶升螺钉与底座齐平后，松开底座两侧共 12 个槽孔螺钉。 - 使用中心千斤顶均匀升降，使送料推料器在主轴后端上下居中；拧紧 12 个螺钉并复查水平。 - 让推料器分别在主轴后端和夹头后端居中，反复调整棒料机后部和前部，直到两个位置均对中。 - 升起上料台，手动将直棒料推入主轴，确认前后均无卡滞、刮擦。 - 用 4 个锚固螺栓固定设备；不得锚固在退让导轨中心位置。锚固后重新复查水平和对中。 - 接通电源后执行上料台校准，再装载棒料进行最终送料验证。
安全要求 ppe_and_safety	- 遵循安全隔离 SOP - 吊装/搬运时人员不得停留在悬吊物或车辆作业区附近
工具与物料 tools_and_materials	- 鱼雷水平仪 - 最大规格送料推料器 - 中心千斤顶 12 个底座槽孔螺钉 4 个锚固螺栓 - 直棒料
验收标准 acceptance_criteria	- 推料器在主轴后端和夹头后端均居中 - 直棒料手动推入时无卡滞/刮擦 - 锚固后水平和对中复查合格 - 上料台校准完成且自动送料无异常阻力
回滚与升级 rollback_or_escalation	无法对中、棒料持续刮擦或设备移动后仍反复触发定位错误时停止送料并联系 LNS。
来源 evidence.page / evidence.section	手册 P10 • SETUP, ALIGNMENT, AND ANCHOR PROCEDURE • 置信度 high
原文摘录 evidence.evidence_text	power off/no pneumatic pressure; align pusher; anchor with four bolts; calibrate loading table

ql80s2_sop_pusher_changeover 送料推料器更换	
触发条件 trigger	棒料直径范围变化或现有送料推料器磨损/损坏
前置条件 preconditions	- 棒料机开关 ON - 上料台处于下位 - 推料器位于机械原点 - 棒料机处于 STOP 模式
作业步骤 steps.action / steps.safety	- 前移上料台，打开主检修盖。 - 将推料器适配器锁止/解锁指向前压并顺时针旋转约四分之一圈，保持在前锁止位置；向前拉推料器但不要完全从载车抽出。 - 将前导向轴承上的白色衬套向棒料机后方拉出，取出推料器。 - 推出横向销，拆下旧推料器适配器；将销重新放回推料器本体以防遗失。 - 将适配器装到新推料器本体，并装上对应白色衬套。 - 先从棒料机前端插入推料器，再将适配器滑入载车；逆时针旋转指约四分之一圈锁止，确认适配器与载车右侧齐平并推料器完全锁定。 - 将载车滑回原点，关闭主检修盖。
安全要求 ppe_and_safety	- 按安全隔离要求操作 - 打开主检修盖前确认设备不会自动启动
工具与物料 tools_and_materials	- 对应直径送料推料器 - 对应白色导向衬套 - 推料器适配器和横向销
验收标准 acceptance_criteria	- 推料器与载车完全锁定 - 适配器/衬套安装方向正确且右侧齐平 - 载车可回到机械原点 - 关闭主检修盖后无安全盖报警，手动送料无卡滞
回滚与升级 rollback_or_escalation	无法锁定、衬套不匹配或推料器与主轴/衬套干涉时停止操作，核对变更组件和主轴衬套规格。
来源 evidence.page / evidence.section	手册 P14 • PUSHER CHANGEOVER PROCEDURE • 置信度 high
原文摘录 evidence.evidence_text	conditions and eight-step pusher replacement procedure

ql80s2_sop_belt_tension 送料载车皮带张力调整	
触发条件 trigger	报警11/12、回零异常、载车运动异常或定期点检发现皮带松弛
前置条件 preconditions	• 棒料机开关 ON • 上料台下位 • 推料器在机械原点 • 棒料机 STOP 模式
作业步骤 steps.action / steps.safety	1. 打开主检修盖并确认载车在原点。 2. 在直线导轨中部测量皮带相对于导轨的垂直位移，不得施加外力。 3. 将位移与约 20 mm 目标值比较；必要时使用棒料机前端张紧器调整。 4. 关闭主检修盖，执行回零和手动低速移动验证。
安全要求 ppe_and_safety	• 打开主检修盖前确认 STOP 状态 • 测量时不得强行压迫皮带
工具与物料 tools_and_materials	• 长度/间隙测量工具 • 皮带张紧器
验收标准 acceptance_criteria	• 皮带中部垂直位移约 20 mm • 载车能够回到原点 • 手动移动无异常阻力和跳齿 • 报警11/12不再出现
回滚与升级 rollback_or_escalation	无法达到约 20 mm、皮带损伤或调整后仍定位异常时停止设备并升级机械检修。
来源 evidence.page / evidence.section	手册 P16 • BELT TENSION ADJUSTMENT • 置信度 high
原文摘录 evidence.evidence_text	measure belt displacement at rail middle; approximately 20 mm; adjust front tensioner

ql80s2_sop_sq5_adjustment SQ5 载车原点接近开关调整

触发条件 trigger	报警12、回零失败或载车轻微前移时 SQ5 未释放
前置条件 preconditions	• 棒料机开关 ON • STOP 模式 • 推料器在机械原点
作业步骤 steps.action / steps.safety	1. 打开主检修盖，确认载车在原点。 2. 调整 SQ5，使其刚好获得稳定信号；目标间隙约 0.5 mm，且不得超过 0.65 mm。 3. 确认载车轻微向前移动时接近开关应释放。 4. 锁紧开关，关闭主检修盖，按遥控器 STOP 清除报警并执行回零验证。
安全要求 ppe_and_safety	• 按安全隔离要求打开主检修盖 • 不得在自动模式下调整
工具与物料 tools_and_materials	• 适用调整工具 • 遥控器
验收标准 acceptance_criteria	• SQ5 稳定检测原点 • 轻微前移可释放 SQ5 • 回零成功且报警不复现
回滚与升级 rollback_or_escalation	无法同时满足稳定检测与前移释放条件时检查开关、支架和皮带，不得扩大间隙强行运行。
来源 evidence.page / evidence.section	手册 P17 • HOME POSITION PROXIMITY SWITCH SQ5 ADJUSTMENT • 置信度 high
原文摘录 evidence.evidence_text	approximately 0.5 mm and no more than 0.65 mm

ql80s2_sop_sq4_adjustment SQ4 直径调整电机接近开关调整

触发条件 trigger	报警16、直径选择动作超时或一次上料两根棒料
前置条件 preconditions	• 料仓无棒料（按手册调整条件） • 上料台下位 • STOP 模式 • 准备两根 6 mm 直棒料用于验证
作业步骤 steps.action / steps.safety	1. 在 Part Setup 中输入棒料直径 6 mm，将两根 6 mm 棒料放在料仓架上。 2. 手动升起上料台；若选择指不动作，先将载车旗标前移到末端再回原点，降台后重复。 3. 若棒料未被上料台拾取，说明开关过高，向下微调并重复动作；上位时棒料应接触台挡块和选择指。 4. 确认单根拾取后，检查设备不会一次装载两根棒料；棒料必须保持直。 5. 按 STOP，恢复实际棒料直径参数并执行手动/自动验证。
安全要求 ppe_and_safety	• 调整期间禁止自动上料 • 确认选择指运动区域无障碍
工具与物料 tools_and_materials	• 两根 6 mm 直棒料 • 遥控器 • 适用调整工具
验收标准 acceptance_criteria	• 6 mm 棒料可正确拾取 • 不会同时拾取两根 • 选择指动作无卡阻 • 实际直径参数下报警16不复现
回滚与升级 rollback_or_escalation	无法稳定单根拾取时停止上料，检查 SQ4、选择指和棒料直线度。
来源 evidence.page / evidence.section	手册 P18 • DIAMETER ADJUSTMENT MOTOR PROXIMITY SWITCH - SQ4 • 置信度 high
原文摘录 evidence.evidence_text	use 6 mm stock; adjust switch until stock loads correctly and does not load two stocks

ql80s2_sop_sq3_adjustment SQ3 测量单元调整	
触发条件 trigger	报警21/30、棒料检测异常或更换测量单元后
前置条件 preconditions	• 料仓无棒料 • 上料台下位 • STOP 模式 • Part Setup 中棒料直径设为 6 mm/最小可用直径 • 使用直棒料
作业步骤 steps.action / steps.safety	1. 手动将上料台升至上位，按 STOP；退回送料机构以获得设备前部操作空间，打开主检修盖。 2. 将直棒料放入通道并从棒料机前端伸出；将测量单元增益顺时针调到最大。 3. 松开固定螺钉，调整测量单元，使红色传感器反射点对准棒料中心，然后锁紧。 4. 将增益逆时针调到最小，再从最小位置缓慢顺时针调节，直到黄色 LED 检测到棒料；检测到后再顺时针增加约四分之一圈。 5. 移出棒料，关闭主检修盖，按遥控器 STOP 清除盖板报警并验证检测。
安全要求 ppe_and_safety	• 调整时不得运行自动循环 • 完成后安装并关闭主检修盖
工具与物料 tools_and_materials	• 6 mm 或最小可用直棒料 • 适用调整工具 • 遥控器
验收标准 acceptance_criteria	• 传感器稳定检测直棒料 • 无棒料时不误报 • 报警21/30条件下检测逻辑符合预期 • 盖板关闭后安全联锁正常
回滚与升级 rollback_or_escalation	LED 无法稳定切换、传感器或线路损坏时停止调整并转电气检修。
来源 evidence.page / evidence.section	手册 P19 • MEASURING CELL ADJUSTMENT - SQ3 • 置信度 high
原文摘录 evidence.evidence_text	center red sensor reflection on bar; adjust gain from minimum until yellow LED detects bar, then add quarter turn

ql80s2_sop_reference 载车原点参考/回零	
触发条件 trigger	编码器丢失原点、报警12、设备移动或断电后需要重新建立位置基准
前置条件 preconditions	• 棒料机通电 • STOP 模式 • 运动区域无人员和障碍
作业步骤 steps.action / steps.safety	1. 从遥控器主菜单进入 MISC. FUNCTIONS，翻页到 REQUEST FOR REFERENCE POINT。 2. 选择 YES，按住 ENTER 约 2–3 秒返回主页。 3. 按 MANUAL MODE 两次，再按 REFERENCE。 4. 确认显示 POSITIONING ERROR / ENCODER LOST ITS REFERENCE POINT 后启动 RESET REFERENCE POINT。 5. 等待载车向原点退回；PLC 等待 SQ5，伺服放大器建立机械止挡扭矩。
安全要求 ppe_and_safety	• 确认上料台和载车可安全运动 • 回零过程中不得接触运动部件
工具与物料 tools_and_materials	• 遥控器
验收标准 acceptance_criteria	• 显示 POSITIONING IN PROGRESS 后完成 • 回到主页且原点位置恢复 • SQ5 输入状态正常 • 手动低速移动无异常阻力
回滚与升级 rollback_or_escalation	回零撞击、无法检测 SQ5 或反复丢失原点时停止设备，检查 SQ5、皮带张力和机械障碍。
来源 evidence.page / evidence.section	手册 P23 • REFERENCE PROCEDURE • 置信度 high
原文摘录 evidence.evidence_text	PLC waits for SQ5 while servo amplifier builds torque at mechanical stop

ql80s2_sop_loading_table_calibration 上料台中心线校准	
触发条件 trigger	更换棒料直径/型材后上料台中心线偏高或偏低，或安装对中后首次校准
前置条件 preconditions	• 料仓有棒料 • 上料台下位 • STOP 模式
作业步骤 steps.action / steps.safety	1. 进入 MANUAL MODE，按 F2 将上料台升至上位并按 STOP。 2. 进入 SERVICE > BAR FEED SETUP，翻页到 NUMBER OF PULSES FOR TABLE CENTERING CALIBRATION。 3. 输入密码 2534 后修改中心线脉冲值；约 1 脉冲对应 0.4 mm 垂直位移。 4. 按 ENTER 确认，按 ESC 退出；按 F3 降台，再按 F2 升台验证修正结果。
安全要求 ppe_and_safety	• 升降动作前确认运动区域无障碍 • 仅授权人员修改校准参数
工具与物料 tools_and_materials	• 遥控器 • 现场棒料
验收标准 acceptance_criteria	• 棒料中心线与主轴入口对齐 • 示例校准量满足 10 脉冲$\approx$上移 4 mm、-5 脉冲$\approx$下移 2 mm 的换算关系 • 升降无障碍且无报警
回滚与升级 rollback_or_escalation	校准后仍偏心或发生刮擦时恢复原参数，重新执行对中锚固 SOP，不得继续自动送料。
来源 evidence.page / evidence.section	手册 P24 • CALIBRATION OF THE LOADING TABLE • 置信度 high
原文摘录 evidence.evidence_text	one pulse approximately 0.4 mm; password 2534; examples 10 and -5 pulses

ql80s2_sop_inspection_cleaning 周期点检与清洁

触发条件 trigger	按维护周期执行，或发现传感器污染、送料异常、油污/粉尘积聚
前置条件 preconditions	• 按安全隔离 SOP 停机 • 确认检修区域安全
作业步骤 steps.action / steps.safety	1. 每日/按现场制度检查急停按钮及安全状态；记录异常。 2. 按手册周期清洁传感器；检查气源过滤单元水位和状态、气动系统密封性等点检项，并记录结果。 3. 外部用软布和清洁产品清洁，内部用布或刷子清洁。 4. 清洁后检查传感器、滚轮、电气柜和防护件，确认无液体残留、异物和松动。 5. 恢复防护件，执行安全联锁和手动送料验证。
安全要求 ppe_and_safety	• 拆盖前断电并关闭气源 • 清洁剂不得接触滚轮、合成材料和电气部件 • 不得使用压缩空气清洁 • 不得使用丙酮、稀释剂等溶剂
工具与物料 tools_and_materials	• 软布 • 刷子 • 适用于设备表面的清洁产品 • 必要的点检记录表
验收标准 acceptance_criteria	• 传感器表面洁净 • 电气部件无水/油/清洁剂接触 • 无压缩空气造成的颗粒滞留 • 防护件和安全联锁正常 • 点检记录完整
回滚与升级 rollback_or_escalation	发现电气进液、传感器损坏、气动泄漏或急停失效时禁止复机并升级维修。
来源 evidence.page / evidence.section	手册 P90 • INSPECTIONS • 置信度 high
原文摘录 evidence.evidence_text	periodic inspections table; clean sensors and bar feeder; check air filtering/pneumatic items

ql80s2_sop_plc_battery PLC 电池更换	
触发条件 trigger	报警46 PLC battery low 或预防性维护到期
前置条件 preconditions	• 按手册条件棒料机通电；由合格人员执行 • 准备同型号电池 • 记录当前 PLC/Parts Library 参数和版本
作业步骤 steps.action / steps.safety	1. 打开电气柜内 PLC 盖。 2. 从标记为 BATT 的插槽取出电池支架。 3. 用同型号电池替换，确认极性方向正确。 4. 将电池支架装回 BATT 插槽，关闭 PLC 盖和电气柜。 5. 检查 PLC RUN、参数、Parts Library 和报警状态，执行受控复机。
安全要求 ppe_and_safety	• 仅合格电气人员可打开电气柜 • 不得给 PLC 电池充电 • 遵循现场 ESD 和电气安全制度
工具与物料 tools_and_materials	• 同型号替换电池 • 适用电气工具 • 参数/版本记录表
验收标准 acceptance_criteria	• 电池型号和方向正确 • PLC 正常运行且参数保持 • 报警46消除 • 受控手动送料验证通过
回滚与升级 rollback_or_escalation	更换后 PLC 参数丢失、无法 RUN 或报警持续时停止设备并联系 LNS/合格电气人员。
来源 evidence.page / evidence.section	手册 P31 • PLC BATTERY REPLACEMENT • 置信度 high
原文摘录 evidence.evidence_text	replace battery with identical type; check direction; BATT bay

sop-001 Controller monthly safety function inspection	
执行频次 frequency	Once a month
触发条件 trigger	Periodic maintenance
前置条件 preconditions	• Qualified personnel • robot work area safe
作业步骤 steps	1. Test teach pendant emergency stop 2. test free drive mode 3. test backdrive mode 4. verify safety settings 5. test active safety inputs/outputs
安全要求 safety_notice	• Stop robot program and disconnect hazardous tool power before repair work
验收标准 acceptance_criteria	• Emergency stop powers off joints • free drive holds position • backdrive releases brakes as expected • safety IO can be triggered
来源 source_section	手册章节 2.1.1,2.1.2

sop-002 Controller visual inspection	
执行频次 frequency	Monthly / every six months / yearly by checklist
触发条件 trigger	Periodic maintenance or abnormal controller condition
前置条件 preconditions	• Disconnect controller power cord
作业步骤 steps	1. Check safety control board terminals and wires 2. check motherboard and safety board connections 3. check controller dirt/dust 4. clean with ESD-safe vacuum if needed
安全要求 safety_notice	• Disconnect power cord before opening controller
验收标准 acceptance_criteria	• Terminals inserted, wires intact, connections secure, no abnormal dust/dirt
来源 source_section	手册章节 2.1.3

sop-003 Clean and replace controller filter	
执行频次 frequency	Once a year or as needed
触发条件 trigger	Dirty filter or periodic maintenance
前置条件 preconditions	• Controller stopped and safe to service
作业步骤 steps	1. Remove filters on both controller sides 2. clean with low-pressure air 3. replace filter if needed 4. gently remove outer plastic frame and maintain filter
安全要求 safety_notice	• Avoid damaging filter frame and controller
验收标准 acceptance_criteria	• Filter clean and seated correctly
来源 source_section	手册章节 2.1.4

sop-004 Robot arm visual inspection	
执行频次 frequency	Every six months / after serious collision for marked items
触发条件 trigger	Periodic maintenance, collision, visible damage
前置条件 preconditions	• Move arm to origin if possible • turn off controller and disconnect power cord
作业步骤 steps	1. Check controller-arm cable damage 2. inspect rubber rings 3. inspect joint lids for cracks/damage 4. check lid screws and tighten to 0.5 Nm
安全要求 safety_notice	• Contact dealer if damage found during warranty period
验收标准 acceptance_criteria	• Cables intact, rubber rings and joint lids undamaged, lid screws tightened to 0.5 Nm
来源 source_section	手册章节 2.2.1,2.2.3

sop-005 Robot arm functional inspection	
执行频次 frequency	Every six months / after serious collision for marked items
触发条件 trigger	Periodic maintenance or collision
前置条件 preconditions	• Correct torque wrench available
作业步骤 steps	1. Check robot arm mounting bolts, tool mounting bolts and screws/bolts connecting joints with torque wrench
安全要求 safety_notice	• Use torque values in section 3.2.4 and user manual mounting specs
验收标准 acceptance_criteria	• Bolts/screws not loose and torque matches specification
来源 source_section	手册章节 2.2.2,3.2.4

sop-006 Manual joint brake release	
执行频次 frequency	As needed
触发条件 trigger	Need to move powered-off robot or package robot without power
前置条件 preconditions	• Power disconnected • dangerous parts not disassembled • mechanical support for base/shoulder/elbow • personnel clear of robot arm
作业步骤 steps	1. Disconnect power supply 2. remove joint lid 3. lift brake pin upward and hold 4. turn joint 5. replace joint lid and tighten screw to 0.5 Nm 6. connect power supply
安全要求 safety_notice	• Do not rotate joints more than 160 degrees; support robot before releasing large-joint brakes
验收标准 acceptance_criteria	• Joint moves safely and lid restored to 0.5 Nm
来源 source_section	手册章节 3.2.2

sop-007 General joint disassembly	
执行频次 frequency	As needed
触发条件 trigger	Joint/link replacement
前置条件 preconditions	• Read manual • prepare repair kit, torque tools, anti-static wristband, rubber rings, M3/M4 hex wrenches, beveled pliers, nylon cable ties • robot supported if removed
作业步骤 steps	1. Move arm to easy disassembly pose 2. turn off power 3. remove joint lid 4. cut cable tie and unplug wire without damaging PCB 5. move rubber ring and supporting ring 6. loosen screws 7. gently separate joints
安全要求 safety_notice	• Use ESD precautions and mechanical support
验收标准 acceptance_criteria	• Joint separated without wire/PCB/rubber-ring damage
来源 source_section	手册章节 3.2.3

sop-008 General joint assembly	
执行频次 frequency	As needed
触发条件 trigger	After joint/link replacement
前置条件 preconditions	• Correct replacement joint and torque tools ready
作业步骤 steps	1. Thread wire bundle through joint 2. align locating pin marks 3. push joints together 4. tighten screws crosswise to specified torque 5. reinstall supporting/rubber rings 6. set white-button flash count to joint number 7. connect and tie wires 8. install lid screws to 0.5 Nm
安全要求 safety_notice	• If flash count is incorrect hold white button and release after correct flashes to reset
验收标准 acceptance_criteria	• Joint number flash count correct, wires tied, torque meets spec, lid secured
来源 source_section	手册章节 3.2.3,3.2.4

sop-009 Replace Safe IO Board	
执行频次 frequency	As needed
触发条件 trigger	Safe IO board fault/replacement
前置条件 preconditions	• Controller powered down completely
作业步骤 steps	1. Turn off controller and disconnect power 2. open cabinet and lay flat 3. remove four flange nylon lock nuts 4. remove/disconnect fan cable, flash card, teach pendant cable, IO plug and fuses 5. remove connector mounting plate screws and grounds 6. remove sheet metal module 7. disconnect power, robot arm and brake cables 8. remove 22 M3 screws 9. replace Safe IO 10. reconnect and restore module 11. power on and test through teach pendant
安全要求 safety_notice	• Ensure controller fully powered down before replacing internal components
验收标准 acceptance_criteria	• Teach pendant test confirms normal function
来源 source_section	手册章节 3.3.1

sop-010 Replace Brake Board	
执行频次 frequency	As needed
触发条件 trigger	Brake board fault/replacement
前置条件 preconditions	• Controller safe to open and power disconnected
作业步骤 steps	1. Lay controller flat 2. release four anti-loosen nuts 3. disconnect fan and teach pendant cables 4. remove sheet metal module 5. disconnect brake board cables 6. remove four cross slot screws 7. replace brake board 8. reconnect wires 9. reinstall module and cables 10. power on and test through teach pendant
安全要求 safety notice	• Avoid flash card/fuse interference with box; keep controller powered off during replacement
验收标准 acceptance_criteria	• Teach pendant test confirms normal brake-board related function
来源 source_section	手册章节 3.3.2

sop-011 Robot packaging	
执行频次 frequency	Before shipment
触发条件 trigger	Robot/controller shipment
前置条件 preconditions	• External tools and electrical connections removed • third-party products safely uninstalled if possible
作业步骤 steps	1. Move robot to packaging pose via JOG: A1 45°, A2 -180°, A3 160°, A4 -70°, A5 180°, A6 -90° 2. shut down 3. disconnect main power 4. disconnect arm from controller 5. load arm and controller into specified box with arm centered
安全要求 safety_notice	• If robot cannot run or power unavailable release brakes manually per section 3.2.2
验收标准 acceptance_criteria	• Robot arm/controller/teach pendant packed responsibly in original Elite Robots package
来源 source_section	手册章节 6